Automated Detection and Mapping of Supraglacial Lakes Using Machine Learning From ICESat-2 and Sentinel-2 Data

Guoping Zhang, Ming Gao, Shuai Xing, Ruiyao Kong, Mofan Dai, Pengcheng Li, Dandi Wang, and Qing Xu

***Abstract*—Although the supraglacial lakes are significant for a comprehensive understanding of the glacier material balance mechanism, obtaining accurate supraglacial lake data by satellite remote sensing is challenging. The lack of 3-D information limits the accuracy of image-based methods, while the Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2) data are limited to satellite trajectories. In this study, the detection and mapping of the supraglacial lakes are implemented by jointly processing the ICESat-2 and Sentinel-2 data. The lake-region images are extracted with the help of ICESat-2 photons, and the lake depths are retrieved. The depths of four supraglacial lakes on the Amery Ice Shelf (AIS), Antarctica, were mapped by trained machine learning methods and Bayesian optimization (BO). The performance of the proposed method is evaluated by comparing it with manual results or self-verification. The results show that joint lake region extraction accuracy is 95.16%. The proposed photon extraction method, combined with BO-categorical boosting (CatBoost), achieves the highest accuracy in retrieval, with a root mean square error (RMSE) of 0.17 m. This study enables automatic operation without input parameters, thus improving the performance and automation level of supraglacial lake observation. With more abundant ICESat-2 data and multispectral images, the proposed method can automatically obtain large-scale surface melting in time and space.**

***Index Terms*—Amery Ice Shelf (AIS), depth retrieval, Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2), machine learning, Sentinel-2, supraglacial lake.**

I. Introduction

ICE mass loss is one of the main factors leading to sea-level rise [1]. Antarctica and Greenland contribute 14.0 ± 2.0 and 13.7 ± 1.1 mm, respectively, to the sea-level rise over the past 45 years [2], [3], and there is an accelerating trend. The increasing supraglacial lakes, melt ponds, and other surface meltings mainly cause this loss. However, observing the surface meltings and evaluating their impact on the Antarctic mass balance is still difficult. This drawback is not only because of the rapid change of the supraglacial lake but also because of its complicated interaction with the Antarctic ice shelf. Specifically, the supraglacial lakes serve as an important reservoir for latent heat [4], whose spectral reflectance is lower than bare ice [5]. Therefore, supraglacial lakes can absorb more heat, creating positive feedback and enhancing surface melting. In addition, the infiltration of surface meltwater into the ice sheets will alter the basal water pressure [6], [7], thereby accelerating the ice flow velocity [8], [9]. With climate warming, supraglacial lakes have become a standard feature at the edges of ice sheets, and their range and duration continue to increase [10], [11]. With the continuous expansion of the supraglacial lake into the ice sheet's interior [10], [12], it is challenging to avoid significant fractures in glaciers and ice sheets [13]. Therefore, monitoring the spatial distribution and depth of supraglacial lakes is essential to understanding the mass balance of ice sheets and controlling the global sea-level rise [14].

As melt precursors and concentrators [15], early knowledge about supraglacial lakes was mainly obtained through field measurements [16], [17]. The researchers reported on the periodic surface melting phenomena on the margins of Antarctica and Greenland [18] and hypothesized the lubricating effect of supraglacial lakes on glacier flow [8], [19]. The drawback of field measurements is evident, as it can only know the surface melting in the local site and a single phase [20]. Limited in situ data have left the time variation of supraglacial lakes unresolved.

Using satellite remote sensing for polar observation has high efficiency and no impact on the ecological environment. However, there are few observations for supraglacial lakes, and it is difficult to fully understand such a small phenomenon by a single observation method. Compared with satellite observation of sea ice, ice sheet, or large-scale atmosphere, Vornberger and Bindschadler [21] compared a Landsat Thematic Mapper (TM) image with an airborne SAR image in 1992 and reported that optical images from the satellite can detect supraglacial lakes. Joughin et al. [22] observed the supraglacial lakes on Greenland only in 1996 using satellite SAR images, while Fair et al. [23] first used the Ice, Cloud, and Elevation Satellite-2 (ICESat-2) to carry

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out light detection and ranging (LiDAR) observation from the satellite.

With the gradual increase of satellite observation of supraglacial lakes, the drawbacks of single data sources are also increasing. Due to the apparent difference in microwave reflection intensity between the supraglacial lake and its surroundings, SAR images have significant advantages in detecting supraglacial lakes. However, water is primarily opaque to microwave frequencies, making retrieval of lake depth impossible. Optical images consider both detection and mapping, but without in situ data correction, images cannot provide the necessary accuracy [24]. Voosen [25] founded that the ICESat-2 blue–green laser can penetrate the water, which makes it possible to measure the depth of a supraglacial lake. However, it can only obtain data along the satellite trajectory [26]. Datta and Wouters [27] used LiDAR data to constrain the empirical estimation of satellite images, including Landsat-8, Sentinel-2, and Planet Labs PlanetScope and SkySat, and jointly retrieved the depths of several supraglacial lakes in Greenland. In the initial attempt, the joint retrieval results of ICESat-2 and satellite images are close to the aerial data but still have limitations. The ICESat-2 data processing method, called Watta, is preliminary. The minimal smoothing in Watta is sensitive to outliers. In addition, although the exponential decay model proves that the statistical model can retrieve the supraglacial lake depth, the performance of more complex machine learning models in estimating the lake depth remains to be discussed.

Therefore, this article attempts to combine ICESat-2 with multispectral images to solve the key problems in the automatic detection and mapping of supraglacial lakes. The 3-D information introduced by ICESat-2 can promote the observation of supraglacial lakes in the real spaceborne remote sensing era. Therefore, the ICESat-2 data and Sentinel-2 images are combined, and supraglacial lakes are automatically detected and mapped by using only spaceborne data through machine learning.

1) The raw data are divided into ice and lake by using statistical indicators, and the $NDWI_{ice}$ range corresponding to the lake pixels is determined by the extracted lake photons, thus extracting the supraglacial lake regions.

2) The processed ICESat-2 photon depths are taken as a priori information, which is matched with the preprocessed Sentinel-2 images. The depth of four supraglacial lakes in the Amery Ice Shelf (AIS), Antarctica, is retrieved using machine learning models and Bayesian optimization (BO).

3) The position and depth of the supraglacial lakes are evaluated and compared with the manual results or test samples, and the performance of the proposed method is analyzed.

II. Related Works

In recent years, SAR, optical sensors, and LiDAR have been installed on spaceborne platforms. Satellite-based observations provide an opportunity to understand the vast area and long-term changes in supraglacial lakes. Due to the limited technology of multisource data joint processing for lake monitoring, the efforts of different remote sensing methods in supraglacial lake observation are summarized.

A. SAR

As an active remote sensing technology, microwaves emitted by SAR can penetrate clouds and observe glacier melt. Microwaves emitted by radar satellites are specularly reflected on the surface of ice lakes, which makes the lakes recorded by SAR images appear as a uniform area with low brightness [28]. In addition, the microwave's penetration ability enables it to detect supraglacial lakes buried under snow or slightly frozen at extreme nights, which has advantages in lake detection [29]. However, the early TerraSAR-X and ENVISAT are unsuitable for observing changing glacier hydrological activities because of their limitations in time resolution or band coverage [30], [31]. Some researchers use Sentinel-1 and GaoFen-3 images and image segmentation to extract weakly scattering objects and identify supraglacial lakes [32], [33]. However, the characteristics of glaciers, such as surface humidity, roughness, and conductivity, also significantly affect the backscattering coefficient of SAR [34], [35], resulting in apparent differences between SAR images acquired in different periods or regions. SAR-based methods pay more attention to large lakes but lack automation in identifying small lakes [36]. Moreover, Sentinel-1 and GaoFen-3 work in the C-band, and their microwaves cannot penetrate the water, so SAR images cannot be directly used to measure the lake depths.

B. Optical Sensor

The reflection, absorption, and scattering characteristics of the supraglacial lakes are obviously different from those of the surrounding snow and ice, which can penetrate the water body and provide water depth information [37]. Therefore, passive remote sensing images can be used for both supraglacial lake detection and mapping.

The detection of supraglacial lakes and pixel extraction are studied comprehensively. The edge detection method considers that the pixel reflectance around the supraglacial lake changes significantly, and the Canny edge operator is used to extract the outline of the ice lake [38]. This image segmentation method takes advantage of different reflectivities of the supraglacial lakes and the surrounding snow when extracting them. The supraglacial lakes recorded in IKONOS images were extracted using the ratio of blue and red bands [39]. To better identify the features of lakes, the normalized water difference index (NDWI) was introduced, and different threshold calculation strategies have been proposed [40], [41], [42]. These methods have been proven to identify lakes in Landsat, MODIS, and Sentinel-2 images. However, image segmentation methods are influenced by clouds and shadows on snow [43]. Neither the object-oriented methods [44] nor the deep learning methods [45], [46] can fundamentally solve the limitations of supraglacial lake detection. Factors, such as moist slush and blue ice, have similar spectral reflectance to melt waters, making the blurred difference between snow and lakes.

The supraglacial lake depth is the key to understand the impact of ice sheet contribution on sea level [47]. However, it is not simple to describe the relationship between image reflectance and depth by radiative transfer equation (RTE) [48]. Assuming that the lakes have a homogeneous bed, the

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difference in reflectance is only related to the depth change. The physical-based bathymetry successfully uses multispectral images with different resolutions (e.g., IKONOS, ASTER, MODIS, Landsat, Sentinel, and WorldView) [43], [49], [50], [51], [52]. However, supraglacial lake beds are not always homogeneous, and the differences in albedo and attenuation parameters of lake beds bring extra errors to the depth retrieval [16], [51], [53]. Researchers use statistical models to construct the relationship between water depth and reflectance and retrieve lake depth in complex environments [43], [53]. However, different passive sensors challenge the robustness of statistical-based retrieval, and the significant demand for in situ data when solving model parameters also limits its use cases.

C. Light Detection and Ranging

LiDAR equipped with a blue–green laser has high-measurement accuracy underwater [54] and can directly obtain the depth of supraglacial lakes [23], [55], [56]. However, airborne bathymetric LiDAR is more expensive and inferior to image retrieval in the measurement range.

The ICESat-2 was launched in September 2018 [57], providing a novel lake detection and mapping means. Although the primary purpose of spaceborne laser altimetry is to monitor the global ice sheet, sea ice, and canopy height [58], it also shows advantages in supraglacial lake monitoring. First, the ICESat-2 measurement range covers 88° north and south latitudes [59], which can cover most of Antarctica and Greenland ice sheets. Second, ICESat-2 is equipped with the Advanced Topographic Laser Altimeter System (ATLAS), and its high-sensitivity photon-counting detectors significantly improve data density. The laser pulses emitted by ATLAS form footprints with a diameter of ~11 m, while the distance between laser footprints along the track direction is only 0.7 m [60]. After processing, the accuracy of ICESat-2 in ice sheet interiors is better than 5 cm [61]. Fricker et al. [23] pointed out that the optical root mean square error (RMSE) of ICESat-2 supraglacial lake depth measurement can reach 0.12 m, while Melling et al. [52] proved that the ICESat-2 data are highly consistent with the ArcticDEM of the same period. Finally, ICESat-2 uses a 532-nm laser, which can measure the accurate lake bed up to ~40 m [62]. However, the angle of the ATLAS laser is fixed, and the raw photons collected by ICESat-2 are distributed in a 2-D profile in the along-track direction. This characteristic makes the joint use of multispectral images and ICESat-2 inevitable [27] and also puts forward higher requirements for ICESat-2 data processing.

The efforts on ICESat-2 data processing are not enough. Currently, the methods of removing noise from ICESat-2 data are primarily designed from the perspective of photon density, which leads users to design different filter kernel shapes or repeat experiments to determine the appropriate threshold according to the environmental changes. ATL13-melt.v1 [63] is necessary to know the location of the supraglacial lake in advance, but it was not easy to know the location ahead. The Watta method [27] is an automatic method developed for supraglacial lakes. Watta estimated the photon elevations over the lake surface, lake bottom, and below the lake using the photon kernel density estimation and identified the location of supraglacial lakes by density and slope, subsequently extracting the lake surface and bed photons. The surface removal and robust fit (SuRRF) algorithm [64] removes noise photons using elevation histograms, with fitting lines identifying the lake surface and bed photons. Although SuRRF is capable of detecting individual lakes, its efficacy diminishes when encountering multiple lakes or uneven lake surfaces within the same dataset. The automated location and depth (ALD) retrieval method [65] removes noise photons through the kernel density estimation, which depends on the threshold setting and is also sensitive to outliers. There are also noise removal methods to analyze the photon height distribution. The lake surface-bed separation (LSBS) [23] removes outliers according to the photon number in the elevation bin. This practice is relatively simple, and it is also necessary to know the location of the ice lake in advance. The adapted ATL08 [66] adjusts the photon extraction method for the ground and canopy to supraglacial lakes. This method first performs a statistical analysis of photon spacing and generates a bimodal Gaussian histogram. In the histogram, the photon spacing corresponding to the intersection of the two peaks is set as a threshold to extract signal photons. Subsequently, these signal photons undergo low-pass filtering to eliminate the effects of topographical undulations. Finally, the photon elevation statistics processed by median filtering enable the differentiation between photons of the lake surface and the lake bottom. However, the adapted ATL08 still requires manual assistance in detecting the position of lakes.

It is generally necessary to know the location of the supraglacial lake in advance to extract the signal photons from ICESat-2 and retrieve the lake depth. Although the ICESat-2 data lack accurate land cover labels [67], most studies on supraglacial lakes ignore this. Xiao et al. [65] detected lake photons according to the hard altitude threshold, which may miss small lakes. Users need to repeat experiments to determine the most appropriate threshold. Subsequently, Flat Lake and Underlying Ice Detection (FLUID) [26] assumes that the supraglacial lake surface is flat to detect the supraglacial lakes automatically, and its performance would decline with the increase in noise photons around the lake.

Therefore, existing satellite observation technologies are often only proficient in supraglacial lake extraction or depth retrieval, and the lack of automation in current methods also limits satellite observations of supraglacial lakes. This study hopes to build upon the existing research to integrate active and passive satellite data for detecting and mapping supraglacial lakes. A workflow that is not sensitive to parameters to automatically process data from ICESat-2 and Sentinel-2 would be proposed, providing technical support for large-scale observation of surface melting phenomena and promoting the advancement of supraglacial lake remote sensing into the satellite era.

III. Study Site and Data

A. Study Site

The study site was selected in the AIS (location: 69°45′S, 71°00′E) in East Antarctica. Compared with the larger Ross

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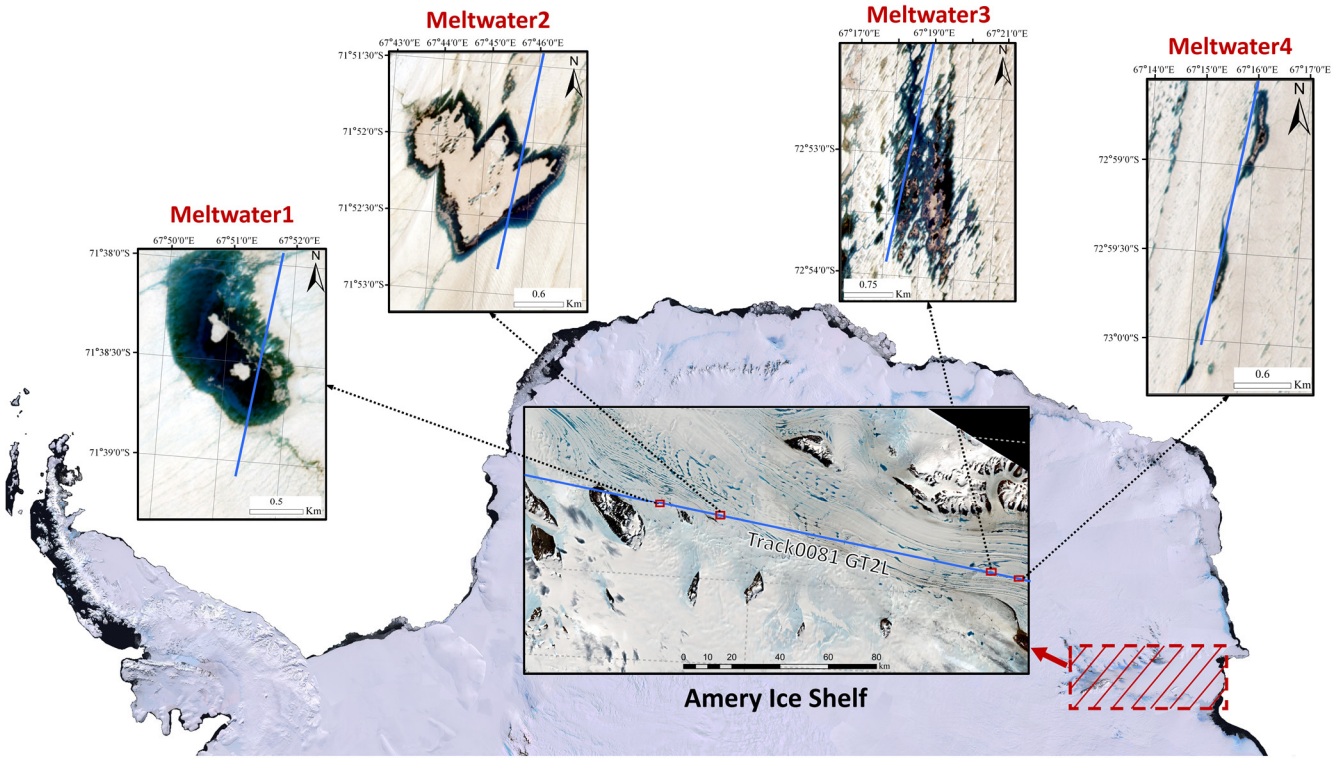


Fig. 1. Study site in AIS. The red dashed box shows the approximate location, while the red boxes mark the locations of the supraglacial lakes, and the blue line represents the ICESat-2 laser track. The background was from Landsat Image Mosaic of Antarctica (LIMA) [79].

and Ronne-Filchner Ice Shelf, the elevation of AIS is lower, and the location is far enough north, so it will experience intense surface melting every melting season [68]. Extensive summer melting was first recorded in 1960 [69], and the largest supraglacial lake can be as long as ~80 km [70]. The hydrological network formed by surface meltwater makes AIS the main estuary for Lambert Glacier to export to the Southern Ocean [71]. As the third largest ice shelf in Antarctica, AIS plays an essential role in the material balance of the ice shelf and the Prydz Bay circulation [72]. As shown in Fig. 1, the red dashed box indicates the approximate location of the study site, the red box indicates the location of supraglacial lakes, and the lakes are numbered from west to east. In contrast, the blue line indicates the ICESat-2 laser track.

B. ICESat-2 ATL03 Data

ICESat-2 is superior to other laser altimeters in data coverage, accuracy, and density, while its use of 532-nm laser beams enables it to measure the depth of meltwaters. In addition, ICESat-2 ATLAS has a total of six laser beams, which are divided into three pairs based on laser energy. The energy ratio between the strong and weak beams is approximately 4:1. The beams are labeled gt1r, gt1l, gt2r, gt2l, gt3r, and gt3l in order of the satellite's orbit direction and form tracks on the surface with an intrabeam spacing of 90 m and interbeam spacing of ~3.3 km [67].

However, ICESat-2 does not provide supraglacial lake data, and Level-3 products, such as ATL07 (sea ice elevation) and ATL10 (sea ice freeboard), often exclude lake bed photons of supraglacial lakes and fail to capture information during the melting season [73]. Therefore, the supraglacial lake study would have to rely on Level-2 ATL03 (geolocated photons) data. ATL03 data contain information, such as the photon time, longitude, latitude, elevation, and other auxiliary parameters. While a large number of noise photons and uncorrected photon elevations make it challenging to apply this data directly, the authors have developed methods in surface classification [74], signal photon extraction [75], [76], scattering signal elimination [77], and refraction correction [78] for bathymetric photons fortunately. These methods can support supraglacial lake detection from ATL03 data and will be introduced in detail in Section III.

In this study, we used ATL03 data collected at the study site on January 2, 2019, which has also been used in other studies [23], [64], [65]. To ensure the laser's penetration for lakes, only the strong beam data (in this case, gt2l) was selected, and its trajectory is shown in Fig. 1.

C. Sentinel-2 L1C Imagery

ICESat-2 data are distributed only along the satellite's orbit, while satellite imagery, such as Sentinel-2, provides an opportunity to extend supraglacial lake observation to a spatial extent. Sentinel-2 consists of two satellites, Sentinel-2A and Sentinel-2B, which were launched by European Space Agency (ESA) in 2015 and 2017, respectively [80]. Although Sentinel-2 is primarily used for monitoring forests, water quality, and land cover in the regions between 56°S and 84°N, its multispectral instrument (MSI) also provides images of Antarctic coastal regions, Baffin Bay, and North Atlantic.

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Sentinel-2 offers a variety of data products, among which Level-1C (L1C) product demonstrates advantages in observing supraglacial lakes [50], [81]. First, the preprocessing is simple. The L1C product provides orthorectified top-of-atmosphere (TOA) reflectance and can be atmospherically corrected using Sen2Cor (open access, version 2.4). Second, the images contain 13 spectral bands ranging from visible, near infrared, and short-wave infrared, which can effectively describe the position and depth variations of supraglacial lakes. Finally, L1C product balances wide-swath coverage and high resolution. The image swath is approximately 100 × 100 km, and the spatial resolution varies among 10, 20, and 60 m depending on the spectral band.

Due to the temporal changes in supraglacial lakes, we not only selected Sentinel-2 L1C images with cloud cover of less than 10% but also aimed for the shortest time interval between the image acquisition and the ICESat-2 ATL03 data. As shown in Fig. 1, the image was also collected on January 2, 2019.

IV. Methods

A. Overview

The method takes the ICESat-2 ATL03 data and Sentinel-2 L1C image as input and automatically implements the detection and mapping of the supraglacial lakes. As shown in Fig. 2, the method is divided into three parts.

1) *Preprocessing:* Signal photons are extracted from the raw data, and the influence caused by the atmosphere and sun glint in the image is corrected so that the reflectance can describe the surface reflection.
2) *Detection:* The photon data are divided into data segments along the ICESat-2 track direction; the photon statistical features in these segments are calculated, and supraglacial lake photons are extracted based on this. The image $NDWI_{ice}$ is calculated and corresponds to the photon coordinates; the $NDWI_{ice}$ range of the lake is determined during the process so that the supraglacial lake region is delineated from the image.
3) *Survey:* Lake photons are classified into lake surface photons and bed photons according to the elevation, the photons generated by laser scattering are eliminated, and the refraction of the lake photons is corrected to obtain the supraglacial lake depth measured by ICESat-2. The photon depths are correlated with image reflectance, and the empirical relationship between photon depth and pixel values is established through machine learning models. The parameters of these models are optimized and subsequently applied to retrieving supraglacial lakes.

B. Data Preprocessing

The ICESat-2 ATL03 data and Sentinel-2 L1C image need to be preprocessed. There are many noise photons in the ATL03 data, so extracting signal photons from Antarctic supraglacial lakes can be difficult. Due to the influence of surface coverage and input parameters, users need to make extra efforts to debug the methods, and the advantages of isolation methods in automation are expected to improve this situation. Unlike the density-based methods, isolation-based methods assume that signal photons are more difficult to isolate or separate than noise photons. Pre-pruning quadtree isolation (PQI) does not need input parameters or neighborhood design. It is adapted to different signal-to-noise ratios (SNRs) and terrain flux [75] and introduced as a typical isolation-based method.

As shown in Fig. 3(a), the signal photons include surface lake bottom and scattered photons caused by lake bed ice. During preprocessing, PQI transforms photons into different quadtree levels according to the photon spatial density, avoids the excessive photon isolation through the prepruning step, generates a frequency histogram of photon elevation and isolation level, and uses the Otsu method [82] to calculate the quadtree level threshold in different elevation ranges, to completely extracting the signal photons of the supraglacial lakes and its surrounding areas.

Specifically, quadtree isolation means that PQI iteratively divides the photon space into four identical subspaces to isolate photons from each other until each subspace contains one photon or no photon. In Fig. 3(b), the signal photon marked by the blue dotted line has undergone three spatial divisions before being separated. If the division process corresponds to the growth of a quadtree, the result of photon division can be reflected by the depth of the quadtree. As shown in Fig. 3(c), the signal photon marked with a blue dotted line is located in the third layer of the quadtree. Therefore, the quadtree depth can reflect the difficulty of photon isolation.

The prepruning step is introduced to optimize the quadtree structure. Prepruning can judge whether the current spatial division is excessive by checking whether the photons are further isolated before and after spatial division. If the photons before and after division remain in the same subspace, they will not be further isolated. Because further spatial division cannot distinguish the noise photons marked by the blue dotted line in Fig. 3(d), the spatial division is stopped in advance. As shown in Fig. 3(e), the prepruning step makes noise photons and signal photons at different quadtree depths. Due to the absorption and scattering of laser pulses by water, the photon density at the lake bottom will decrease with the increase in water depth. The prepruning step is advantageous in a low-SNR environment, as it helps identifying underwater signals accurately.

Therefore, by comparing the differences of photons in the quadtree depth, signal photons can be extracted. The Otsu method has the advantage of maximizing the quadtree depth variance of signal and noise and is used to calculate the threshold of quadtree depth D. The Otsu method is used to calculate the depth variance σ^2 between different types of photons, as shown in the following equation:

$$\sigma^2 = \omega_s(d)\left(\mu_s(d) - \mu(d)\right)^2 + \omega_n(d)\left(\mu_n(d) - \mu(d)\right)^2 \quad (1)$$

where σ^2 is the interclass variance, $\omega_s(t)$ is the proportion of potential signal photons, $\omega_n(d)$ is the proportion of potential noise photons, $\mu_s(d)$ is the mean depth of signal photons, $\mu_n(d)$ is the mean depth of potential noise photons, and $\mu(d)$ is the mean depth of all photons, as shown in the following

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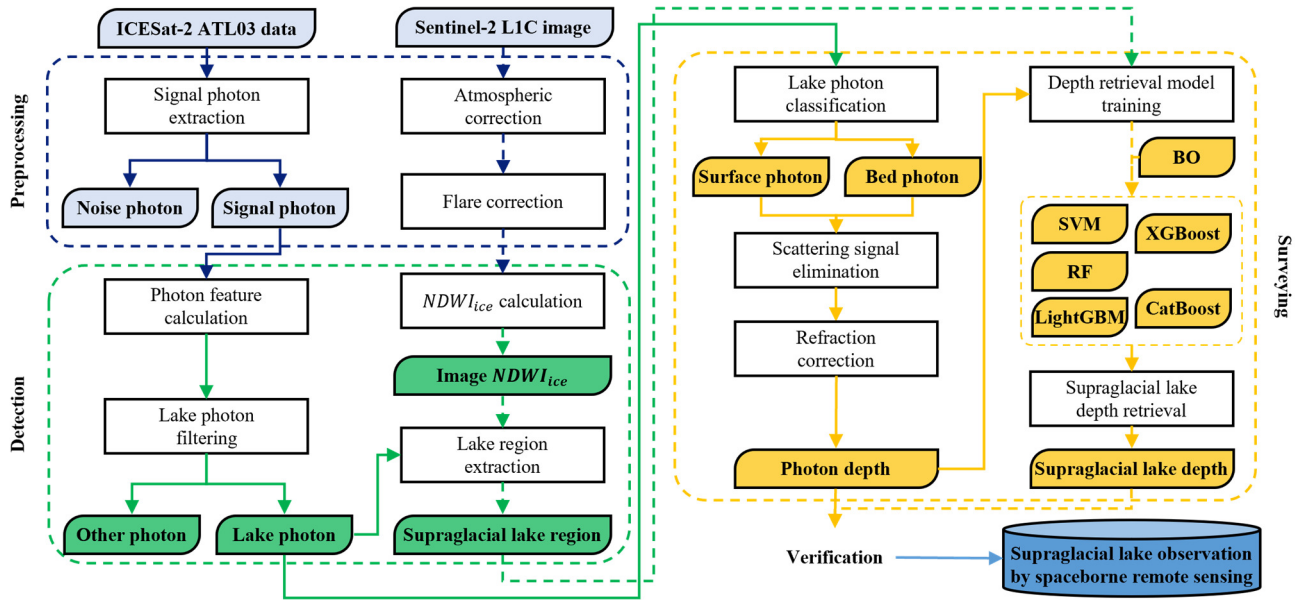


Fig. 2. Flowchart of the supraglacial lake detection and mapping.

equations:

$$\omega_s(d) = \frac{d}{n} \tag{2}$$

$$\omega_n(d) = 1 - \frac{d}{n} \tag{3}$$

$$\mu_s(d) = \frac{\sum_{i=1}^{d} D_i}{d} \tag{4}$$

$$\mu_n(d) = \frac{\sum_{i=d+1}^{n} D_i}{n-d} \tag{5}$$

$$\mu(d) = \omega_s(d)\mu_s(d) + \omega_n(d)\mu_n(d). \tag{6}$$

By changing $d, \mu_s(d)$ and $\mu_n(d)$ are recalculated, and σ^2 is renewed. Find the largest σ^2, the corresponding depth D_{Max} is selected as the threshold, and the photons with an D_i greater than the threshold are extracted as signal photons.

During image preprocessing, the Sentinel-2 L1C image is atmospheric corrected using the ESA Sen2Cor toolbox to reach the L2A level. In addition, the solar reflection of the lake surfaces may lead to flare contamination. The corresponding relationship between the blue and near-infrared bands is established through covariance to find and correct the sun glint in the image [83]. In addition, because the spatial resolution of each band in a Sentinel-2 image is different, they are resampled to 10-m resolution before further processing.

C. Supraglacial Lake Detection

In this step, the statistical properties of signal photons are calculated and divided into lake photons and other photons. These photons not only delimit the ICESat-2 data over supraglacial lakes but also help to determine the NDWI_{ice} range of the supraglacial lake region in the image.

1) Supraglacial Lake Photon Filtering: The difference between the surface topography and its ability to reflect laser pulses changes the statistical features of the ICESat-2 signal photons. The Normalized Photon Rate-Elevation Ratio (NPRER) was designed and applied to the automatic land-sea classification [74]. In the Antarctic supraglacial lake and its surrounding areas, the photon rate and elevation of signal photons are also different. Surface melting has been proven to have a high reflectivity to satellite laser altimetry data, and its specular reflection of ICESat-2 blue–green laser gives the corresponding data bins a higher photon rate than the data covered with ice or snow. ICESat-2 obtains the photons on the lake surface and bed simultaneously so that the signal photons in the supraglacial lake area have a larger elevation standard deviation (STD). Therefore, the Detection Index (DI) was put forward based on NPRER for Antarctic data to describe the difference of photons in the supraglacial lakes and surrounding areas

$$\text{DI}_i = \frac{\text{PR}_i - \text{PR}_{\text{Min}}}{\text{PR}_{\text{Max}} - \text{PR}_{\text{Min}}} \cdot \frac{\text{STD}_i - \text{STD}_{\text{Min}}}{\text{STD}_{\text{Max}} - \text{STD}_{\text{Min}}} \tag{7}$$

where $\text{PR}_{\text{Max}}, \text{PR}_{\text{Min}}$, and PR_i refers to the data bin with the largest, smallest photon rate, and the photon rate of the ith data bin, respectively, while STD_{Max}, STD_{Min}, and STD_i refers to the data bin with the largest, smallest elevation STD, and the STD of the ith data bin, respectively. The STD can be calculated by the following equation:

$$\text{STD} = \sqrt{\frac{\sum_1^n (e_j - \bar{e})^2}{n}} \tag{8}$$

where n is the number of photons in the data bin, and e_j and $\bar{e}$ refers to the elevation of the jth photon and the average elevation of signal photons in data bin, respectively.

Since the supraglacial lake data has a larger DI, the photons corresponding to the peak with a larger DI in histogram can be labeled as lake photons. The Otsu method is also used to calculate the DI threshold, and the data bin with DI higher than this threshold is marked as a supraglacial lake bin, while the corresponding photons are labeled as lake photons.

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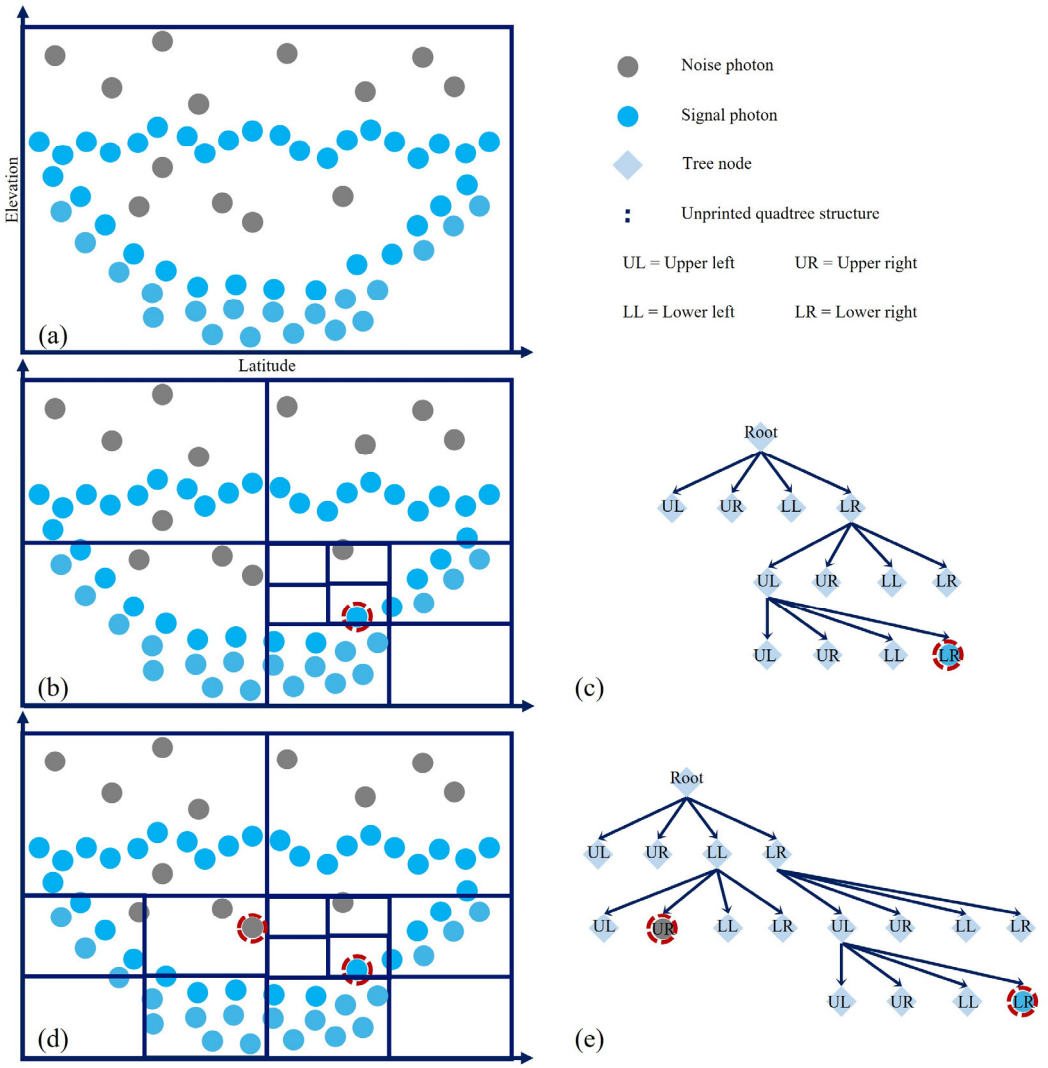


Fig. 3. Principle of PQI. (a) Original state with labeled noise and signal photons. (b) and (c) Isolated photon and corresponding quadtree. (d) and (e) Prepruning identifies noise photon cluster and modifies quadtree structure.

2) Joint Supraglacial Lake Region Extraction: The supraglacial lakes and their surrounding areas have different spectral characteristics, so $\mathrm{NDWI_{ice}}$ can enhance the spectral contrast of open melting and snow. This technique is applied to the satellite image in supraglacial lake detection [84]

$$\mathrm{NDWI_{ice}} = \frac{B_{\mathrm{Blue}} - B_{\mathrm{Red}}}{B_{\mathrm{Blue}} + B_{\mathrm{Red}}} \tag{9}$$

where B_{Blue} and B_{Red} refers to the image reflectance in the blue band and red band, respectively.

Generally, the $\mathrm{NDWI_{ice}}$ of lake pixels is lower than that of surrounding pixels, but $\mathrm{NDWI_{ice}}$ may completely fail in very wet and saturated areas. The lack of spatial information limits the performance improvement of image-based methods. ICESat-2 data directly measure the surface elevation, and the extracted lake photons record the land cover labels and coordinates. Combining the ICESat-2 data with the image can expand the ICESat-2 results from the laser trajectory and extract the lake region more accurately. During the process, the $\mathrm{NDWI_{ice}}$ is calculated. Since the used image and ICESat-2 data were collected on the same day, there does not adopt registration step, but matched the ICESat-2 photon label and the image $\mathrm{NDWI_{ice}}$ according to the coordinates. As shown in (4), the max $\mathrm{NDWI_{ice}}$ of the corresponding pixel of lake photons is taken as the threshold, and the pixels with $\mathrm{NDWI_{ice}}$ less than the threshold are marked as lake pixels, so the supraglacial lake region can be extracted with the help of ICESat-2 data

$$\mathrm{NDWI_{thres}} = \mathrm{Max}\left(\mathrm{NDWI_{ice-}}_{i2}\right) \tag{10}$$

where $\mathrm{NDWI_{ice-}}_{i2}$ is the $\mathrm{NDWI_{ice}}$ of the pixels corresponding to ICESat-2 lake photons and $\mathrm{NDWI_{thres}}$ is the $\mathrm{NDWI_{ice}}$ threshold obtained with the help of ICESat-2 data.

D. Supraglacial Lake Survey

Satellite multispectral images have been used to estimate the supraglacial lake depth based on empirical models, but the lack of in situ depth challenges the empirical model solving. The supraglacial lake image and coincident photons are obtained,

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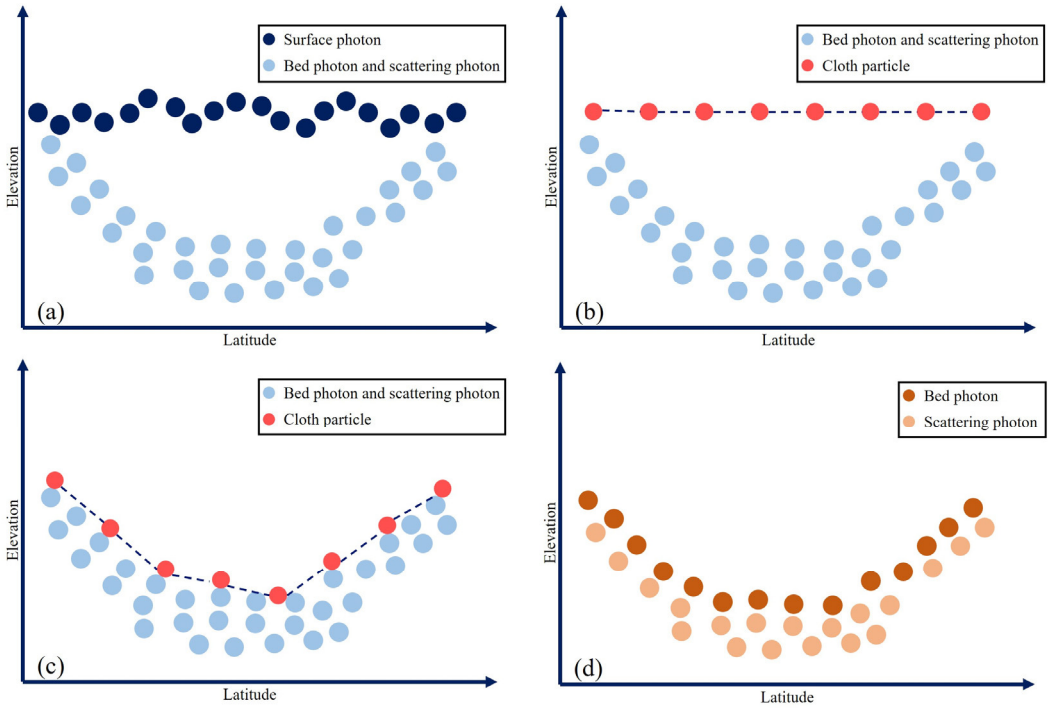


Fig. 4. Illustration of the scattering signal elimination. (a) Result of the lake photon classification. (b) Initial state of the cloth simulation. (c) Final state of the cloth simulation. (d) Result of scattering signal elimination.

and the empirical model training can be implemented by using lake photons instead of in situ measurement. Lake photons are classified into surface photons and lake bed photons, and lake surface scattering photons and ice scattering photons are eliminated using the cloth simulation method. The bathymetric error of lake bed photons caused by refraction is corrected. The photon depths and the corresponding pixel value are used to train the machine learning models, and the depth of the supraglacial lake is retrieved.

1) Photon Depth Calculation of Supraglacial Lake:

a) Lake photon classification: The lake surface photons should first be discriminated against the lake bed photons and scattering photons. In this study, the elevation histogram of the lake photons is calculated and fit by the Gaussian curve. Then, the upper boundary H_{Max} and lower boundary H_{Min} of surface photons are determined with the elevation corresponding to the Gaussian curve peak h_{maxpho} as the center and the width threshold of twofold Gaussian STD (2σ), and the photons with an elevation less than H_{Min} are labeled as lake bed photons, as expressed in (5) [85]

$$\begin{cases} \mathrm{SP} = (\mathrm{SD}_k, \mathrm{SH}_k), & \left(H_{\mathrm{Min}} < \mathrm{SH}_k < H_{\mathrm{Max}}, k = 1, \ldots, N_{\mathrm{sp}}\right) \\ \mathrm{BP} = (\mathrm{BD}_l, \mathrm{BH}_l), & \left(\mathrm{BH}_l < H_{\mathrm{Min}}, l = 1, \ldots, N_{\mathrm{bp}}\right) \end{cases} \tag{11}$$

where SP refers to the surface photons and BP refers to the lake bed photons and scattering photons; SD_k and BD_l refer to the along-track distance of surface photons and bed photons, respectively; SH_k and BH_l refer to the elevation of surface photons and bed photons, respectively; and N_{sp} and N_{bp} are the number of photons in the corresponding category.

b) Scattering signal elimination: When ICESat-2 measures snow and ice, the laser pulses will experience scattering in the subsurface [86]. As shown in Fig. 4(a), the scattered photons generated by the lake bed have a longer optical path than the surface photons, and they are distributed below the actual reflection surface, thus affecting the accuracy of lake depth measurement [87].

The cloth simulation is a classic method introduced to filter scattering photons. Imagine a soft cloth big enough to cover the photons on the lake bed. The final shape of the cloth will be the bottom shape. The simulated cloth is abstracted as a grid structure consisting of cloth particles and springs between them, so the deformation of the cloth is determined by the external and internal forces of cloth particles in the elevation direction. To describe the surface details, Zhang et al. [88] stipulated that the deformation is regulated by hardness and needs to be set according to the terrain fluctuation and the user's prior knowledge. Zhang et al. [77] simplified the cloth simulation and successfully applied it to the ICESat-2 data.

As shown in Fig. 4(b), the cloth particles are evenly distributed on the lake bed, and the distance between them is 10 m. This distance can describe the details of the lake bed without being affected by the noise nearby [77]. The hardness of the cloth is set to be the softest, and under the action of particle gravity and tension between springs, the cloth is deformed. When cloth particles are in contact with photons, they are set to be immovable. As shown in Fig. 4(c), when all the cloth particles do not move, the simulated cloth would be the shape of the lake bed. Photons near the cloth particles will be marked. As shown in Fig. 4(d), these photons are densified using the progressive irregular triangular network to obtain the lake bed photons without scattering.

C. Refraction correction: Refraction correction of lake bed photons is also necessary. The refraction of laser pulses

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by water leads to the shift of the photon position. Our former study uses a simple but accurate method [78]. First, the photon ID and along-track distance recorded in ATL03 data are used to connect the surface and bed photons, and the laser pulse propagation path is recovered. Then, the laser refraction under lake surface fluctuation is classified, and Snell's law implements the refraction correction of photon coordinates. Therefore, using the elevation difference of the lake surface photons and lake bed photons, the supraglacial lake depth on the ICESat-2 track can be obtained. Compared with Parrish et al. [62], our method is able to correspond to surface and bed photons more simply and quickly.

2) Joint Depth Retrieval of Supraglacial Lake: The current research [16], [27] combined photon depth and image reflectance to retrieve supraglacial lake depth, but followed the empirical relationship of exponential decay and faced the challenge of environmental factors such as complicated sediment reflectance. The machine learning development enhances the nonlinear prediction ability, which enables the carrying out of lake surveys without prior information about water reflectance, quality, or sediment. The support vector regression (SVR) with radial basis function (RBF) kernel is used to retrieve lake depth, which is proven to establish a reliable relationship between water depth and image reflectance [89].

In addition, the methods based on the decision tree are nonparametric and straightforward, so random forest (RF), extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), and categorical boosting (CatBoost) are all used to retrieve the depth.

The RF generates the final depth result through the linear combination of several decision trees [90]. By training each tree with bootstrap sampling and feature bagging, RF can introduce randomness to reduce the over-fitting phenomenon of the lake depth results.

The other methods change the weight of training depth samples through a boosting strategy. XGBoost is designed based on the gradient lifting decision tree (GBDT). The second-order Taylor expansion and regularization of the water depth function are expected to improve the bathymetry accuracy and reduce the risk of over-fitting [91].

Like XGBoost, LightGBM is an optimized implementation of GBDT [92]. Both use the negative gradient of the loss function as the residual approximation of the current decision tree and use it to fit the new decision tree. However, LightGBM uses gradient-based one-side sampling (GOSS) for depth samples to obtain the best depth retrieval.

The innovative CatBoost proposed by Yandex Company has been widely used in academic circles [93]. CatBoost introduced target statistics during supraglacial lake depth retrieval, transforming the relationship between lake depth and water color into a new numerical feature. Under the framework of GBDT, CatBoost will randomly arrange the training samples, calculate the gradient under different arrangements, and update the model, thus reducing the deviation of gradient estimation and model prediction and improving the model's generalization ability. The advantages of CatBoost in overcoming gradient bias and solving prediction bias make the performance of depth retrieval can be expected [94].

For the initial parameter settings, the regulation parameter C of SVR is set to 1, and the kernel parameter γ is set to 1×10^{-5}. The initial parameter settings in RF, XGBoost, and LightGBM are identical. The preferred number of decision trees is 100, and the maximum depth of the trees is 6. For CatBoost, the number of iterations is 500, the learning rate is 0.03, and the maximum depth of the trees is 6.

BO is introduced to independently optimize hyperparameters for each model, enabling an objective comparison of their performance [95]. During the computational process, BO's current hyperparameter settings refer to previous evaluation results. The hyperparameter optimization relies on an internal holdout set. The training set of the lakes was further split into an 80% training subset and a 20% validation subset through the random sampling. The validation subset is used to evaluate the performance of different hyperparameter combinations, effectively mitigating the risk of over fitting. By leveraging the Gaussian process as the surrogate model, BO captures the underlying patterns in the hyperparameter space. The expected improvement acquisition function prioritizes hyperparameter combinations that are likely to yield the most significant improvement in the model performance.

Specifically, the initial range for the regularization parameter C of SVR is $(1, 100)$, and the initial range of γ is $(1 \times 10^{-7}, 1)$. The parameter ranges for RF, XGBoost, and LightGBM are consistent: the preferred range for the number of decision trees is $(100, 500)$, and the range for the maximum depth of the trees is $(3, 15)$. For CatBoost, the range for iteration times is $(20, 500)$, the learning rate is $(0.01, 0.15)$, and the range for the maximum depth of the trees is $(3, 15)$. A total of 150 BO iterations were carried out for hyperparameter tuning. Before optimization, each model used 15 random hyperparameter sampling as the initial setting.

E. Verification

The lack of in situ data makes it difficult to verify the accuracy. Since the proposed research pays little attention to the detection of the supraglacial lake, we need to collect the supraglacial lake photons manually and screen the lake pixels from the image so that we can quantitatively evaluate the performance of the proposed detection method. Our manual collection results are uploaded (https://zenodo.org/doi/10.5281/zenodo.10978211).

The lake depth results are verified from both qualitative and quantitative aspects. The depth results were compared with those collected by Fricker et al. [64], [65]. Then, the Watta and adapted ATL08 algorithms (see Section II) were chosen as comparison methods due to their high automation.

In addition, the accuracy and size of photons obtained in different depth ranges and the influence of photon number on subsequent steps are also analyzed to ensure that the best photon depth results are used for joint depth retrieval.

Since it is impossible to collect the complete lake depth, the joint lake depth retrieval accuracy is calculated using the leave-one-lake-out principle. In the leave-one-lake-out principle, one lake serves solely as an independent testing set for accuracy evaluation, with the other three lakes used for model training and hyperparameter optimization. As mentioned above, BO

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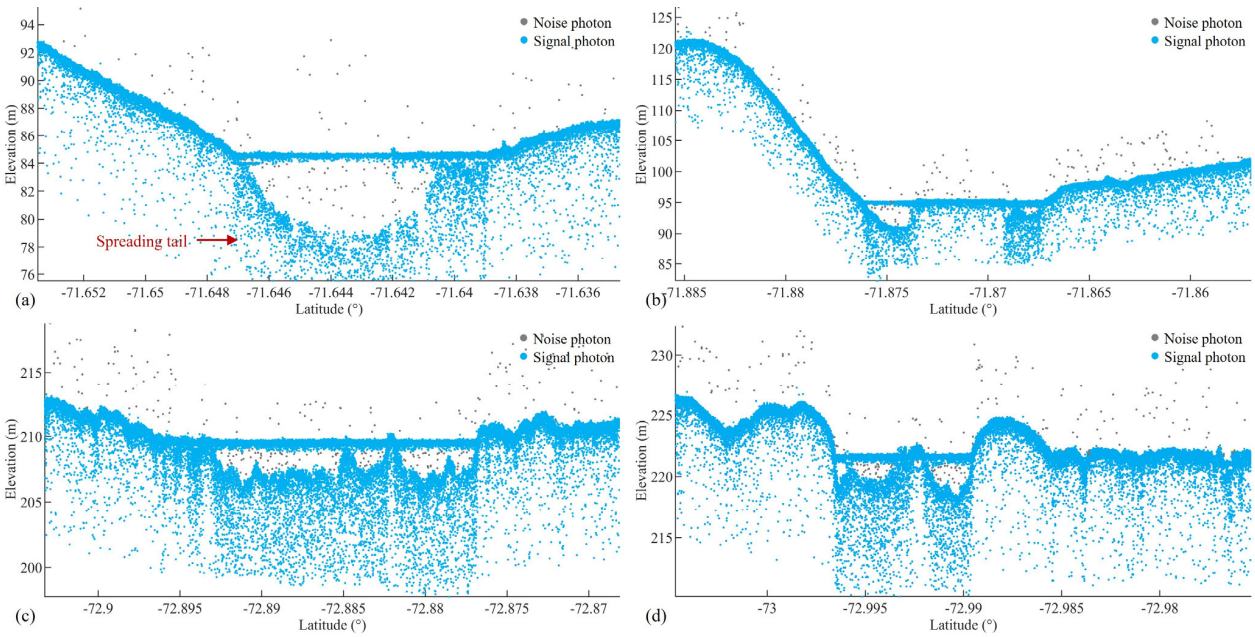


Fig. 5. Signal photon extraction results using PQI. (a)–(d) Result at supraglacial Lakes 1–4. Among them, blue points are the signal photons and gray points are the noise photons. The red arrow indicates the spreading tail phenomenon caused by ice scattering below the lake bed.

further divides the training set using an internal holdout set. Specifically, BO allocates 80% of the samples as the training subset and 20% as the validation subset to obtain the optimal hyperparameter combination. Then, the trained model can be used to retrieve the water depth of the remaining supraglacial lake. The ICESat-2 measurement of that lake can be used to calculate various accuracy indicators, including mean absolute error (MAE), RMSE, and correlation coefficient R^2. This method can avoid the influence of sample correlation on the test in the case of limited supraglacial lakes.

The optimal model would be selected, and its results would be further analyzed. Then, scatterplots of the blue and green band reflectance of the retrieved results would be drawn. The exponential relationship of blue, green, and red bands with depth is calculated [96], which quantitatively shows the feasibility of the exponential decay relationship in the supraglacial lake depth retrieval.

V. Results

A. Signal Photons Extracted From ICESat-2 Data

Fig. 5(a)–(d) corresponds to the extracted results of supraglacial lakes 1–4, respectively. The PQI removes noise photons in the air and water column, which are caused by atmospheric scattering and water scattering. It should be noted that the PQI extracted the photons from the ice, snow, lake surface, and the photons under the lake bed. Photons under the lake bed are often caused by ice scattering, which leads to the spreading tail phenomenon observed from Fig. 5(a) [87]. The previous processing experience shows that dense scattering photons are challenging to eliminate accurately by the noise removal method [65]. These scattered photons are retained in the results and would be eliminated by the cloth simulation to avoid underestimating the elevation of the lake bed.

B. Supraglacial Lake Detection Results

1) Supraglacial Lake Photon Filtering: Fig. 6(a)–(d) shows the lake photon filtering results. In each subfigure, the first and second lines show the results from the top and side views, respectively. The third-to-fifth lines show the change of STD, PR, and DI along the ICESat-2 track direction, respectively.

Compare the filtering results with the data collected manually, and the accuracy is also shown in Fig. 6. It can be found that the filtering method can accurately identify lake photons. The results in shallow-water regions, such as lake boundaries, are satisfactory. The results show the method's effectiveness. The overall accuracy is 99.21%, and the best result (accuracy of 99.58%) appears in Lake 1. Whether the lake photons are continuous (such as Lake 1 and Lake 3) or discontinuous (such as Lake 2 and Lake 4), the accuracy of the results is better than 98.76%.

It is appropriate to use PR and elevation STD to express the statistical difference between lake photons and surrounding photons. Lake depth has a significant impact on STDs. With the increase in depth, the photon height difference between the lake surface and the bottom also increases, leading to increased elevation STD. Because the lake boundary's elevation difference is similar to its surroundings, it is challenging to define the lake boundary only by the STD. The water surface and lake bottom reflect the laser at the lake boundary, which makes the photon number in this area higher than in other areas, making PR capture this change. With the further increase in water depth, laser absorption by water bodies leads to a decrease in the photon number on the lake bed. At this time, PR's ability to detect lakes is limited. Therefore, the DI proposed in this study combines the advantages of elevation STD and PR and further highlights the difference between the supraglacial lake and the surrounding area, thus ensuring the accuracy of supraglacial lake photon filtering.

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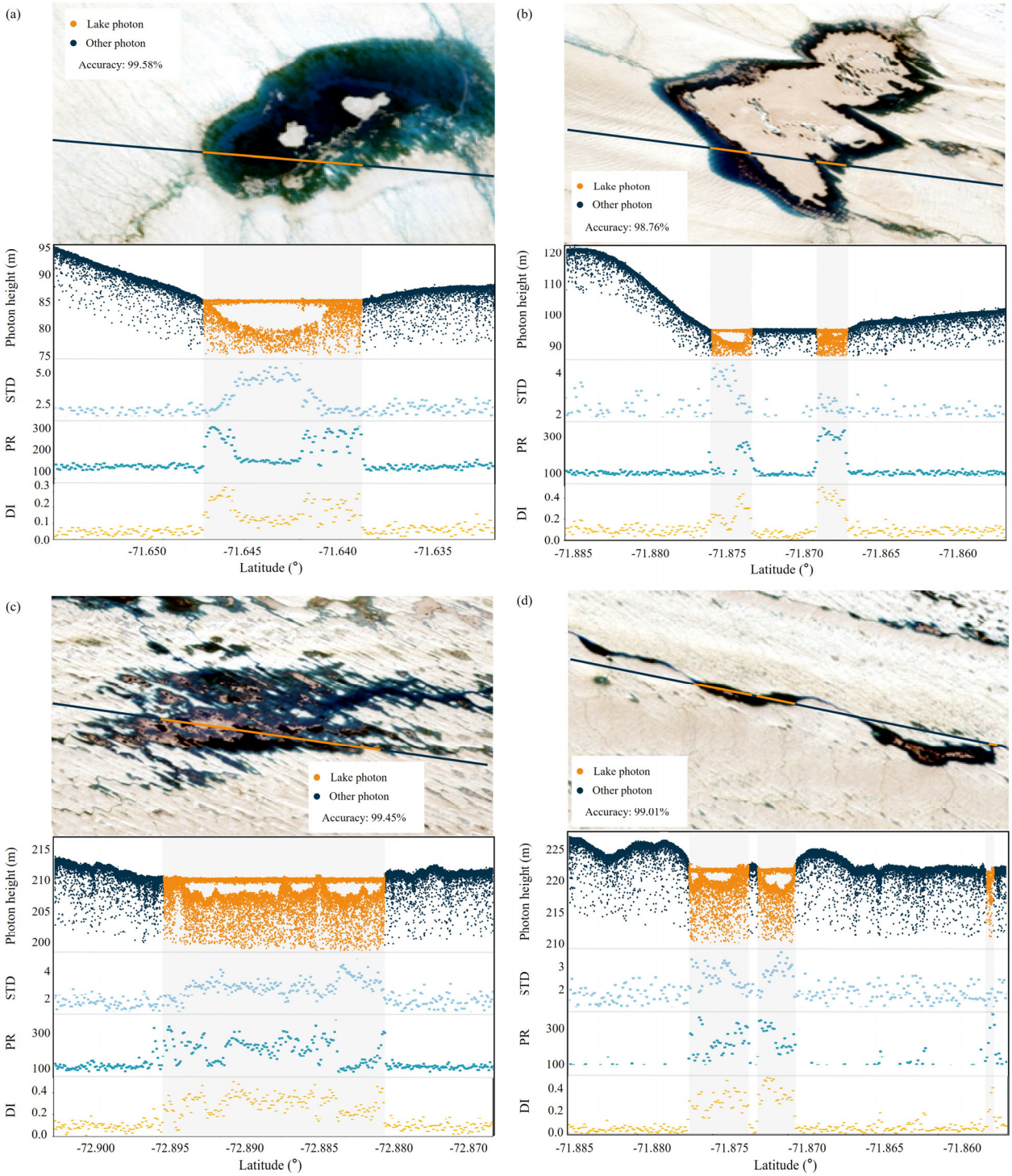


Fig. 6. Supraglacial lake photon extraction results. (a)–(d) Supraglacial Lakes 1–4, respectively. In each subfigure, the topline shows the Sentinel-2 image overlaid with ICESat-2 track; the second line shows the signal photon side view; and the third, fourth, and fifth lines show the corresponding STDs, PRs, and Dis, respectively. Among them, the supraglacial lake photons are painted orange, while other photons are painted dark blue, and orange shadows emphasize the supraglacial lake position.

2) Supraglacial Lake Region Extraction: Fig. 7 shows the supraglacial lake region obtained with photon assistance, and the extraction results are highlighted on the multispectral image using orange. We found that the $NDWI_{ice}$ thresholds of different lakes have no noticeable difference; they are all less than or equal to 0.4. Since the melting stages of supraglacial lakes are similar in the same image, the close $NDWI_{ice}$ threshold indicates that the joint extraction method is effective.

Furthermore, the overall accuracy is 95.16% by comparing the extraction results with the data collected manually, which shows that the joint extraction method not only gets rid of

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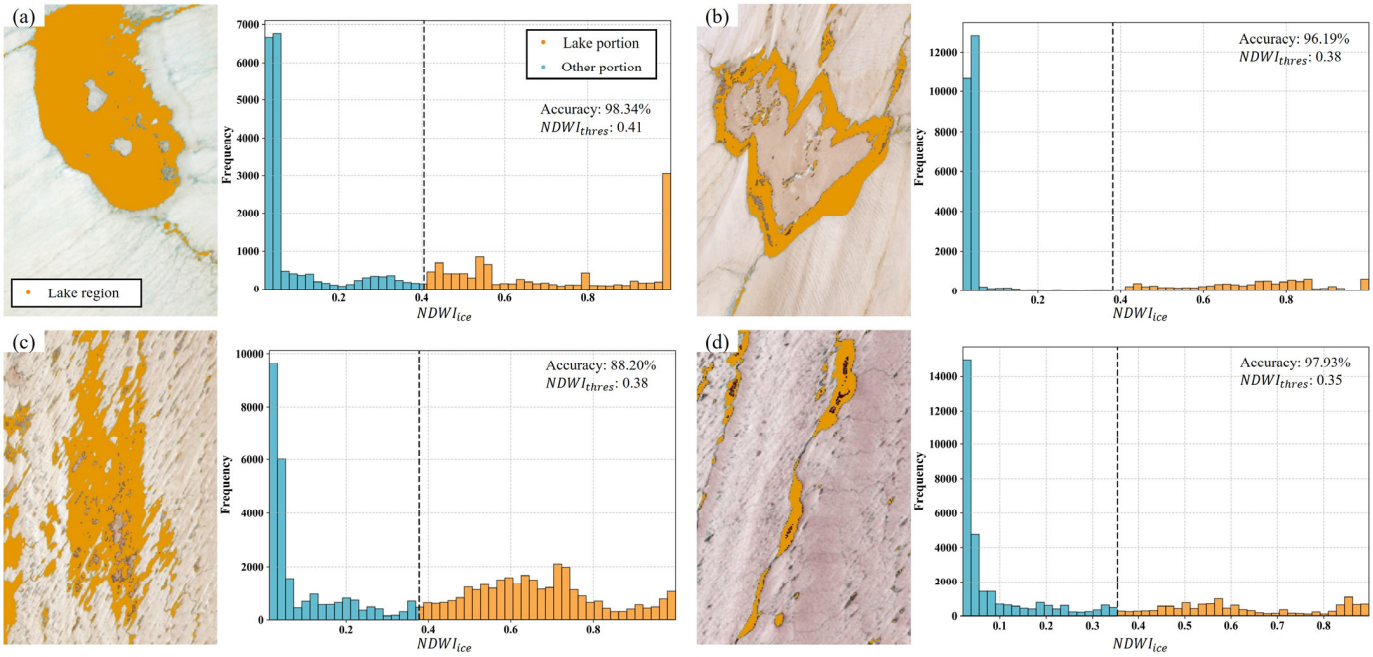


Fig. 7. Extraction results of supraglacial lake region. (a)–(d) Lakes 1–4. In each subfigure, (left) lake region with orange and (right) corresponding $NDWI_{ice}$ histogram and the threshold calculated by the Otsu method.

the complicated calculation process of $NDWI_{ice}$ threshold in previous studies but also has good recognition ability for supraglacial lake pixels with different depths. The spatial distribution of lakes affects the precision of extraction. Supraglacial lakes with continuous spatial distribution (as shown in Lakes 1, 2, and 4) quickly obtain higher accuracy. Among them, the extraction accuracy of Lake 1 reached 98.34%, the highest among the lakes. As shown in Lake 3, it is hard to extract a subdivided lake, with an accuracy of 88.20%. The difference between this subdivided lake pixel and the adjacent surrounding pixels is challenging to explain intuitively.

Further observation of Lake 3's $NDWI_{ice}$ histogram shows an apparent bimodal distribution. Lake 3 $NDWI_{ice}$ obtained by the proposed method is 0.38, which can divide the histogram into the lake part and other part. Therefore, the introduced lake photons make it possible to extract the lake area simply and accurately from images. In particular, the vertical information lake photons provide this method advantages in dealing with the subdivision of supraglacial lakes.

C. Supraglacial Lake Depth Retrieval Results

1) Accuracy of the Lake Photon Depth: During accurate photon depth retrieval, the lake photons are used to classify the lake surface and bed, eliminate scattering signals, and correct refraction errors. Fig. 8 shows the supraglacial depth results. The lake depths obtained by different methods are compared with the manual results of the lake bed.

The depths obtained from RTE are drawn by red lines, which directly use the existing results [64]. RTE has challenges in accurately capturing water's complex scattering and absorption characteristics. In practical application, RTE often significantly underestimates the depth of supraglacial lakes, which shows that it is unrealistic to retrieve water depth only by radiation transmission characteristics and multispectral images.

Based on ICESat-2 data, our proposed method, the Watta, and the adapted ATL08 are compared and represented by the orange, light blue, and green lines. The lake bed retrieved by the Watta fluctuates, and there are many overestimates. For example, the positions of 71.642°S in Lake 1°, 72.886°S in Lake 3, and 72.993°S in Lake 4 are all overestimated. In contrast, the lake bed curves retrieved by the proposed method and adapted ATL08 are closer to the manual results.

Therefore, the ICESat-2 results are further calculated. The scatter plots of these and the manual results are shown in Fig. 9. For the four lakes, the overall RMSEof the adapted ATL08 is the best, and the RMSE of Watta is the lowest. When the adapted ATL08 processes Lake 1, the RMSE is 0.10 m, the highest accuracy among all results. The overall RMSE of the adapted ATL08 is 0.16 m. The overall RMSE of the proposed method is 0.26 m, slightly worse than the adapted ATL08. The overall RMSE of Watta is 0.41 m. In Fig. 8(e)–(h), there are many abnormal values in the lower right of the Watta fitting line, which indicates that Watta's limitation in dealing with abnormal values may lead to an overestimation of the depth.

Accuracy indicators are shown in Table I. Further analysis shows that our proposed method and the adapted ATL08 have their advantages and disadvantages. The proposed method and adapted ATL08 have the highest correlation with manual results in Lake 1, while the overall R^2 of the proposed method is 0.91, and the adjusted ATL08 is 0.96. The R^2 between the retrieved lake depths and the manual results are also significantly different. Watta's R^2 is 0.77, and there are many abnormal points in its scatter plots [see Fig. 8(e)–(h)]. This

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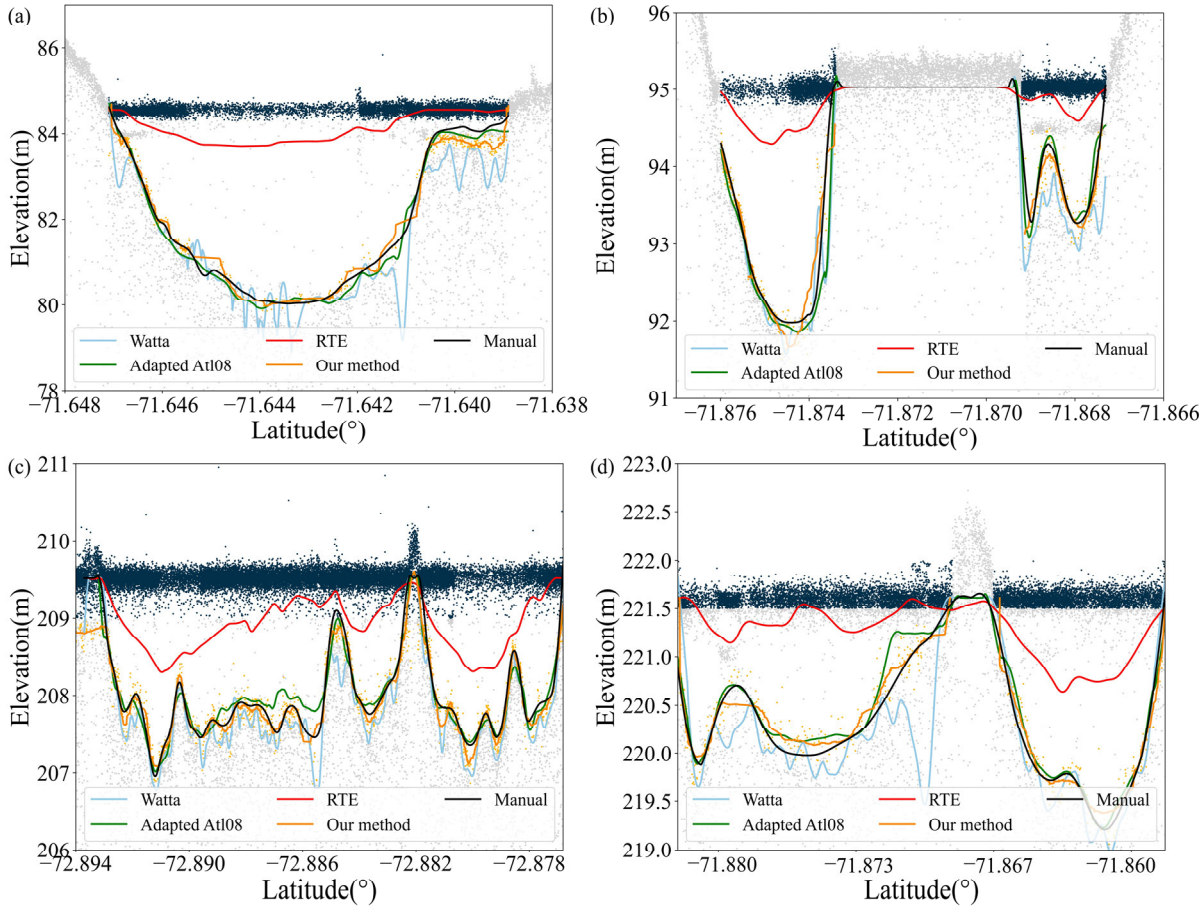


Fig. 8. Lake depth retrieved by different methods. (a)–(d) Lake bed profiles calculated by different methods for Lakes 1–4, light blue line is for Watta, green line is for adapted ATL08, red line is for RTE, orange line is for our proposed method, and dark blue line is for the manual result.

situation is consistent with the conclusion that the lake bed retrieved by Watta fluctuates.

The mean error (ME) is used to understand the deviation of the extracted photons and the manual results. The ME of our proposed method is 0.01 m, and that of the adapted ATL08 is 0.05 m. Therefore, the method proposed in this article has a slight deviation. The average depth of Lake 4 is the smallest, and both methods slightly underestimate the depth at this time. In other cases, the methods often overestimate the depth. Watta's overall ME is 0.22 m, which indicates that this method overestimates the depth of the supraglacial lakes. This finding is consistent with the analysis above.

Dividing the depth at the interval of 1 m, the accuracy of the three methods in different water depth ranges is shown in Fig. 10. The error is the largest within the [0, 1)m range, which indicates that the abnormal values contained in highly shallow water have a significant impact on the performance. In the [1, 5)m, the error increases with the increase of lake depth, which shows that depth is an important factor affecting performance. Watta has the biggest RMSE and the biggest ME in any depth range. The RMSE of the adapted ATL08 in different depth ranges is slightly smaller than the proposed method. In the range of the [1, 3)m, the depth deviation retrieved by the adapted ATL08 and proposed method is minimal. Our proposed method's deviation in retrieving the [3, 4)m range is less than that of the adapted ATL08, but the deviation is the opposite in the [4, 5)m range.

2) Size of the Lake Photon Depth: The number of lake photons also significantly impacts the performance of machine learning. In machine learning, an adequate number of lake photons has the potential to improve the accuracy of supraglacial lake depth retrieval. Therefore, the photons extracted by different methods are also calculated and further analyzed. The lower right of Fig. 9 shows the number of photons extracted to retrieve the curve of the lake bed. Among the four lakes, our proposed method extracts the most photons from the lake bed, while the number of photons extracted by Watta and adapted ATL08 is less. In Lake 2, 2024 photons are extracted by the proposed method, while the adjusted ATL08 extracts 1694 photons, and the difference between them was the smallest. In Lake 3, the proposed method extracts 8106 more photons than the Adapted ATL08. The proposed method uses cloth simulation to extract photons, and the distance between clothed particles is set to 10 m, which means that this method has a better chance of extracting more photons within the same distance.

Fig. 10(c) also shows the number of photons detected by the three methods within different height ranges. As the depth increases, the number of photons detected by the three methods decreases. Due to the photon absorption and scattering by

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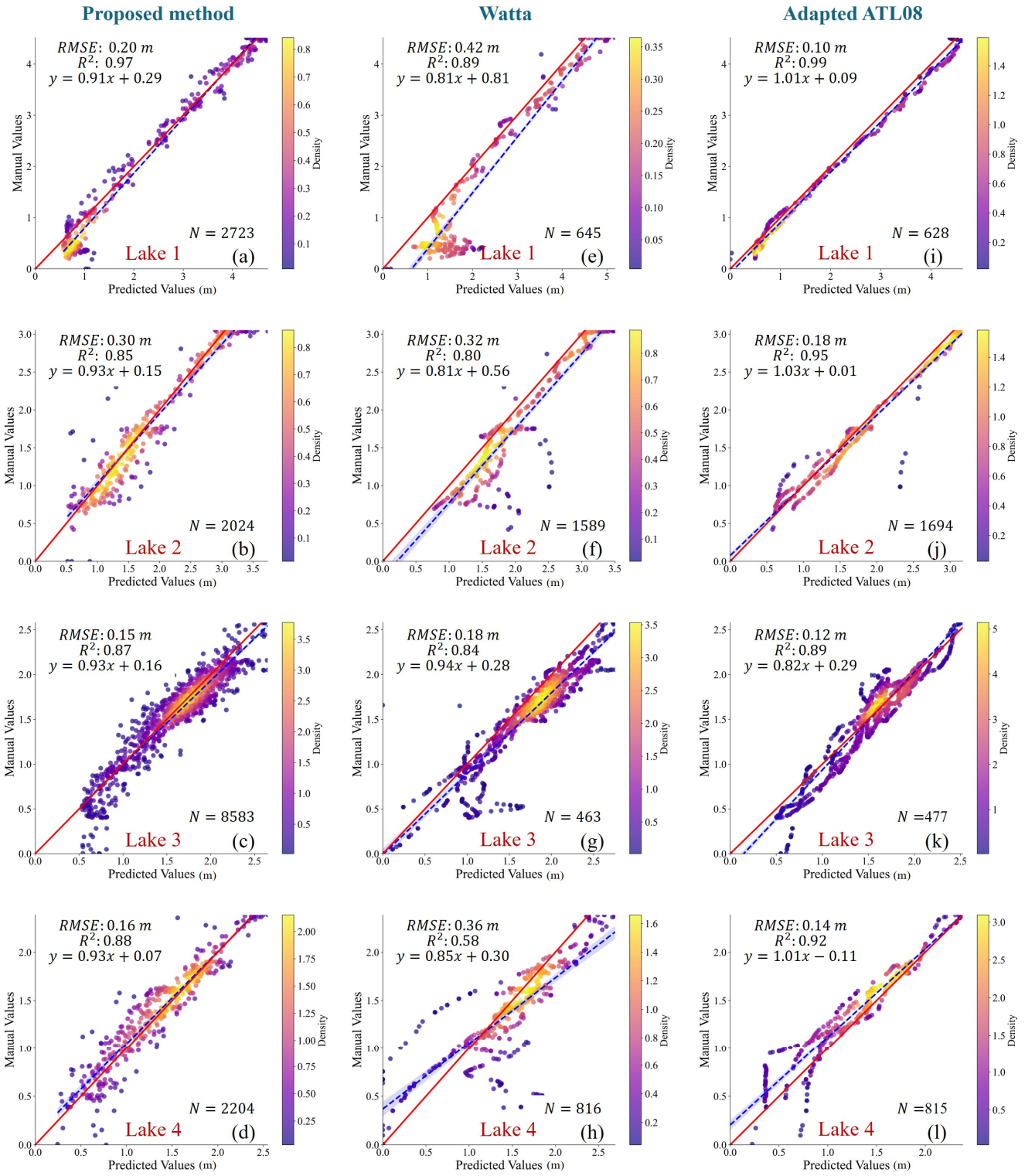


Fig. 9. Lake depth scatterplots using different methods for Lakes 1–4 compared with the manual result. (a)–(d) Our proposed method, (e)–(h) Watta, and (i)–(l) adapted ATL08, the red solid line is 1:1 line, while the purple dotted line is the fitting line.

water bodies, the photon number in the intervals [3, 4) and [4, 5)m is particularly low. Despite variations in photon numbers across different intervals, the number of photons extracted by the proposed method is significantly higher than the other two methods. The number of photons extracted by our proposed method is several times greater than those extracted by Watta and adapted ATL08. These photons can form a larger sample set for in-depth retrieval.

3) Joint Depth Retrieval Results: Different photon results, models, and their parameter optimization strategies have varying impacts on the supraglacial lake depth retrieval accuracy.

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TABLE I
ACCURACY INDICATORS OF PHOTON DEPTH CALCULATION

Lake	Method	Max depth (m)	Mean depth (m)	RMSE (m)	ME (m)	R^2
Lake 1	**Manual**	**4.52**	**2.10**	-	-	-
	Watta	5.29	2.55	0.42	0.45	0.89
	Adapted ATL08	4.66	2.21	0.10	0.17	0.99
	Proposed method	4.76	2.17	0.20	0.11	0.97
Lake 2	**Manual**	**3.05**	**1.77**	-	-	-
	Watta	3.50	2.00	0.32	0.24	0.80
	Adapted ATL08	3.17	1.82	0.18	0.06	0.95
	Proposed method	3.04	1.79	0.30	0.02	0.85
Lake 3	**Manual**	**2.58**	**1.57**	-	-	-
	Watta	2.76	1.75	0.18	0.18	0.84
	Adapted ATL08	2.52	1.58	0.12	0.01	0.89
	Proposed method	2.65	1.62	0.15	0.02	0.87
Lake 4	**Manual**	**2.38**	**1.39**	-	-	-
	Watta	2.69	1.50	0.36	0.10	0.58
	Adapted ATL08	2.38	1.29	0.92	-0.10	0.92
	Proposed method	2.44	1.35	0.16	-0.04	0.88
Overall	**Manual**	-	**1.66**	-	-	-
	Watta	-	1.88	0.41	0.22	0.77
	Adapted ATL08	-	1.67	0.16	0.05	0.96
	Proposed method	-	1.70	0.26	0.01	0.91

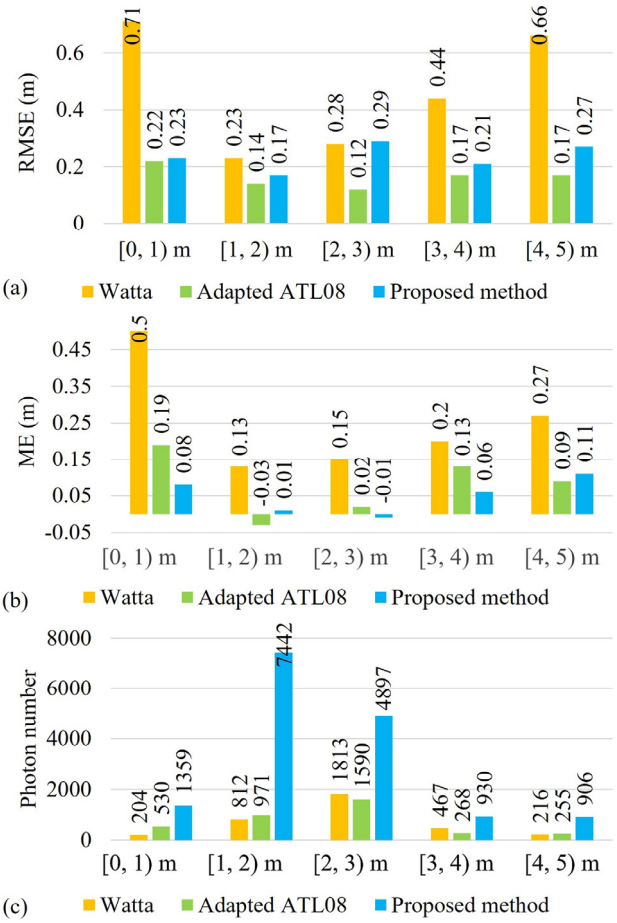


Fig. 10. Accuracy indicators at different depths. (a) RMSE, (b) ME, and (c) photon number. The yellow bar represents Watta, the green bar represents adapted ATL08, and the blue bar represents the proposed method.

The accuracy under different combinations has been verified and will be discussed in Sections VI-C and VI-D. Among these combinations, the photons extracted by our proposed method, combined with BO-CatBoost, and achieved the highest accuracy when mapping four supraglacial lakes.

The model parameters for each lake were determined based on the other three lakes, following the leave-one-lake-out principle. For Lake 1, the model achieved 251 BO-CatBoost iterations, a learning rate of 0.11, and a tree depth of 10, leading to a maximum depth of 6.03 m with an MAE of 0.17 m, an RMSE of 0.30 m, and an R^2 of 0.88. In contrast, Lake 2, with 209 iterations, a learning rate of 0.07, and a tree depth of 10, had a slightly lower maximum depth of 5.29 m but exhibited an MAE of 0.08 m, an RMSE of 0.16 m, and an R^2 of 0.99, reflecting a more accurate retrieval. Lake 3, with 100 iterations, a learning rate of 0.18, and a tree depth of 10, produced a maximum depth of 3.04 m, an MAE of 0.10 m, an RMSE of 0.15 m, and an R^2 of 0.99. Lake 4 had the shallowest maximum depth of 2.65 m, with an MAE of 0.03 m, an RMSE of 0.07 m, and an R^2 of 0.99, achieved after 223 iterations, a learning rate of 0.07, and a tree depth of 4.

Fig. 11 further shows the depth results for each lake calculated by BO-CatBoost, with consistency verified against image reflectivity. By comparing Fig. 11(e) and (h), it can be observed that depth gradually increases from the lake boundary toward the center. The depth retrieval results highlight the varying maximum depths among the lakes, with Lake 1 being the deepest, followed by Lake 2–Lake 4, which is the shallowest.

Fig. 11(i)–(l) depicts the change of reflectivity with depth in different bands. The reflectance of blue, green, and red bands decreases exponentially with the depth increase. The reflectivity of the red band decays the fastest and is entirely absorbed by the water body in the range of ~1.5–3 m. The reflectivity of the blue and green bands has not dropped to 0, indicating that these two bands can penetrate the supraglacial lakes. The depth retrieval ability of bands makes them of different importance in different depths. The rapid change of reflectivity with depth in the red band gives it an advantage in describing the change of shallow-water depth, while the blue and green bands provide a basis for retrieving the change of deep lake bed.

In addition, the exponential curves fit according to the reflectivity are also drawn in Fig. 11(i)–(l). Many points are on both sides of the exponential curve, making it hard to ignore the fitting error. Therefore, the exponential attenuation based on the water depth reflectivity [16] is challenging to establish in retrieving the supraglacial lakes. The discovery explains the apparent difference between the depth obtained by RTE and the photon depth of ICESat-2 in Fig. 8. Although the accuracy of the image-based method in the shallow sea is satisfactory [97], this method is very rough in estimating

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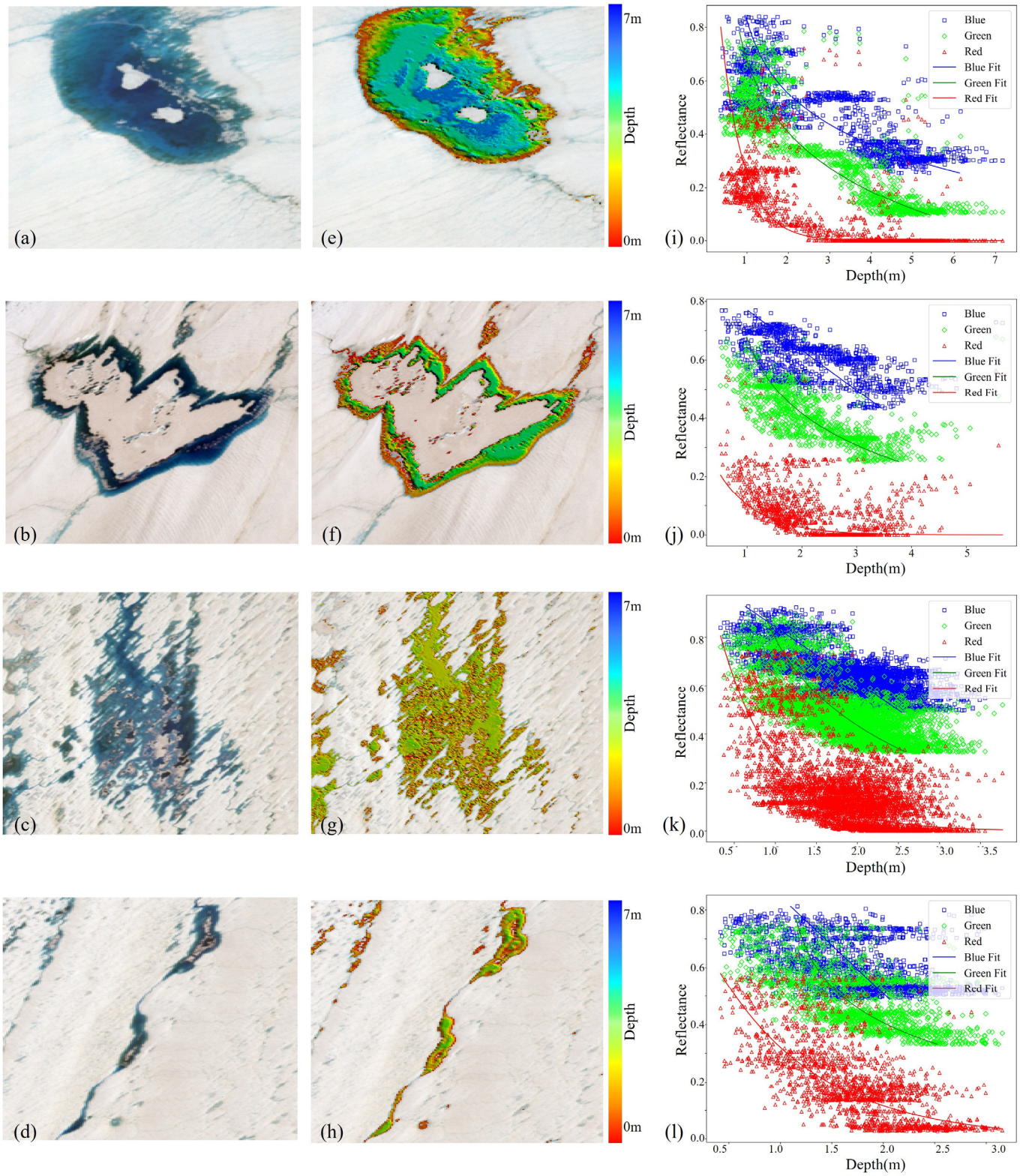


Fig. 11. Supraglacial lake joint depth retrieval results using photons extracted by the proposed method and BO-CatBoost. (a)–(d) Used Sentinel-2 image for Lakes 1–4. (e)–(h) Retrieved water depth of Lakes 1–4, in which the change of pixel color from red to blue represents the depth increase. (i)–(l) Reflectance change of Lakes 1–4 with depth, and the exponential fitting of each band is also drawn.

the supraglacial lakes [17], [98]. The advantage of learning machines in the nonlinear prediction is that it makes it easier to obtain better results than empirical relationships in depth retrieval. Compared with the image-based method, the joint depth retrieval assisted by ICESat-2 data is more reliable and can capture the lake bed details.

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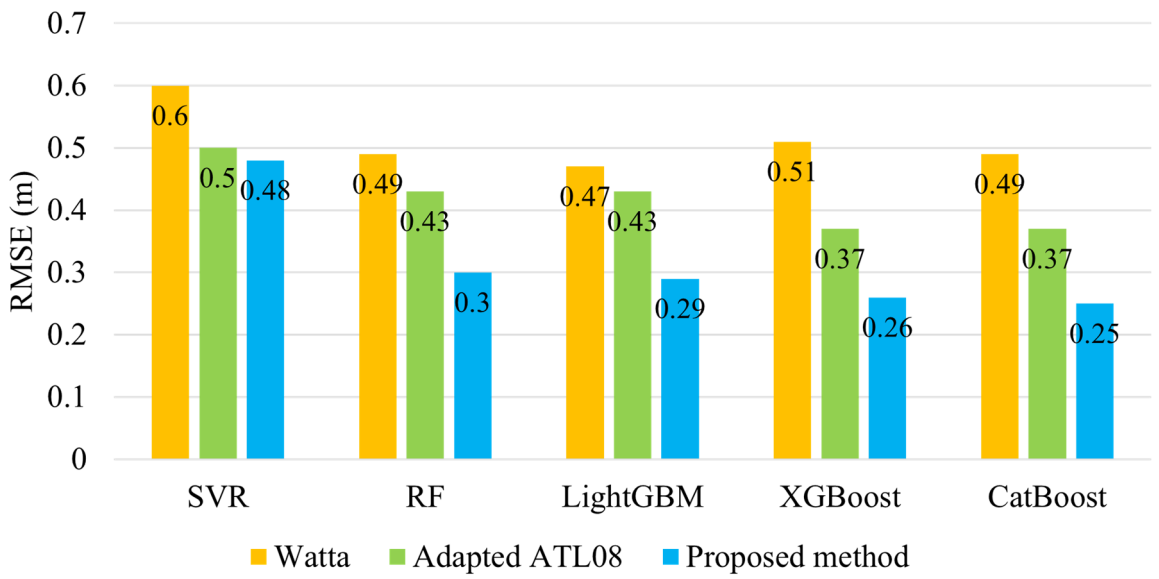


Fig. 12. Retrieval RMSE uses different photon results and initial models before parameter optimization. The yellow bar represents Watta, the green bar represents adapted ATL08, and the blue bar represents the proposed method.

VI. Discussion

Although the best results for supraglacial lakes have been demonstrated, the multifactorial effects on retrieval accuracy remain to be clarified. In this study, the photon results, models, and parameter optimization strategies can be combined to produce 120 different depth outcomes across the four supraglacial lakes, and the performance of these combinations awaits verification and discussion.

A. Impact of the Photon Results on Joint Depth Retrieval

Before optimizing the model parameters, the impact of photon results on the initial models was verified. The results of Watta, the adapted ATL08, and the proposed method were used to train SVR, RF, LightGBM, XGBoost, and CatBoost, respectively. The accuracy of the results was verified using the leave-one-lake-out principle, meaning that photon results from three lakes were used to train the models, while the result from the fourth lake served as the validation set. Fig. 12 shows the mean RMSE of depth outcomes in the study area using different photon results. Here, manually collected lake bed data [64], [65] were used as the reference, while lake bed photons extracted by the Watta method, Adapted ATL08, and our proposed method were used as training labels. The results indicate that the photon results significantly impact the initial models' performance.

Watta's RMSE was the highest when training all initial models. The RMSE of Watta combined with SVR and XGBoost was 0.60 and 0.51 m, respectively. The limitations of Watta in the accuracy and quantity of photon extraction restrict its performance in joint depth retrieval. In contrast, the RMSE of adapted ATL08 gradually decreased with the increasing nonlinearity of the models. The RMSE of adapted ATL08 combined with SVR was 0.50 m, while the RMSE was 0.37 m when combined with either XGBoost or CatBoost. The proposed method demonstrated the lowest RMSE among all initial models. When using the photon results extracted by the proposed method, the maximum RMSE for supraglacial lake depth was 0.48 m, achieved with SVR, while the RMSE was only 0.25 m in combination with CatBoost.

It is important to note that while a more significant number of photons leads to better model performance, this improvement hinges on the precision of the photons. The results of combining Watta with the initial models show that merely increasing the photon quantity without ensuring their quality can introduce noise that degrades model training. Therefore, the superior performance of the proposed method in training models can be attributed to both the increased number of photons and the reduced noise, which enables the model to learn more accurately. This dual advantage—more photons and better-quality photons—explains why our proposed method outperforms the others.

B. Retrieval Improvement by Parameter Optimization

Ideally, BO can tune the model parameters to enhance their generalization ability, thereby compensating for the shortcomings of the photon results and yielding more accurate outcomes. Fig. 13 illustrates the RMSE of retrieved depths using different photon results in combination with models optimized through parameter tuning. Similarly, lake bed photons extracted by different methods were used as training labels, while manually extracted lakebed data were utilized for validation.

The RMSE achieved by Watta remains the worst among the three photon extraction methods, ranging from 0.32 to 0.49 m. When combined with BO-RF, BO-LightGBM, and BO-XGBoost, the RMSE of Watta continues to increase, indicating that Watta has limited adaptability to complex models. The adapted ATL08 has an overall better RMSE than Watta. When the adapted ATL08 is combined with BO-LightGBM, the RMSE is the highest at 0.41 m. When the adapted ATL08 is combined with BO-CatBoost, the RMSE is the lowest at 0.31 m. The proposed method demonstrates a significant

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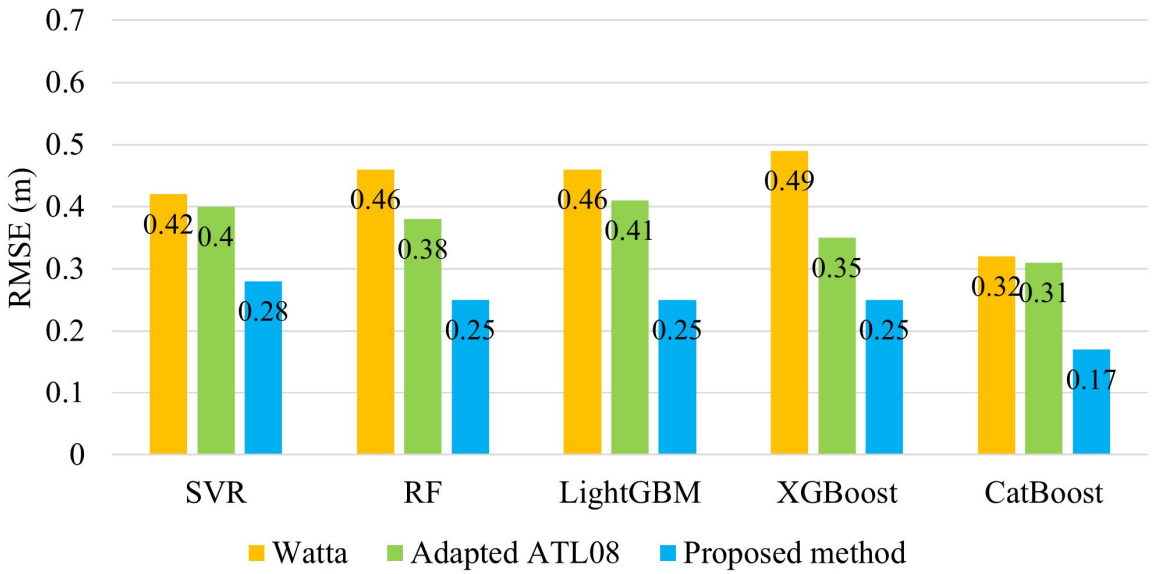


Fig. 13. Retrieval RMSE uses different photon results and models after parameter optimization. The yellow bar represents Watta, the green bar represents adapted ATL08, and the blue bar represents the proposed method.

TABLE II
RELATIVE RMSE IMPROVEMENT OF DIFFERENT COMBINATIONS AFTER PARAMETER OPTIMIZATION

Photon extraction method	Retrieval model	Relative RMSE improvement (%)
Watta	BO-SVR	30.00
	BO-RF	5.83
	BO-LightGBM	1.51
	BO-XGBoost	4.50
	BO-CatBoost	34.55
Adapted ATL08	BO-SVR	19.73
	BO-RF	11.18
	BO-LightGBM	4.76
	BO-XGBoost	4.74
	BO-CatBoost	16.33
Proposed method	BO-SVR	41.27
	BO-RF	17.86
	BO-LightGBM	14.95
	BO-XGBoost	3.71
	BO-CatBoost	32.34

advantage, with an RMSE consistently below 0.28 m. When combined with BO-CatBoost, the RMSE is as low as 0.17 m.

Figs. 12 and 13 show that BO improved all the results to varying degrees. Table II illustrates the relative improvement brought by parameter optimization. The most significant improvement was observed in the combination of the proposed method and SVR, with a relative improvement of 41.27%. The least noticeable improvement was seen in the combination of Watta and LightGBM, with a relative improvement of only 1.51%.

For the photon extraction methods, when combined with the models optimized by BO, Watta achieved a relative improvement of 15.26%, the adapted ATL08 achieved 11.35%, and the proposed method achieved 22.03%. The quantity and accuracy of the photon results affected the degree of optimization by BO. The accuracy of photons extracted by Watta was relatively poor, making it difficult for BO to break through the bottleneck of photon errors. The proposed method extracted the most significant number of lake photons with better accuracy, and BO further highlighted the advantages of high-quality photon results. Therefore, the proposed method is considered the most suitable for joint depth retrieval and will be used in the subsequent steps.

BO enhanced the models' generalization ability at the retrieval model level. The performance of SVR improved on average by 30.31%. However, it was observed that the improvement was more significant when SVR was combined with higher quality photon results, indicating that performance gains cannot rely solely on hyperparameter tuning. The performance of RF increased by 11.62%, LightGBM improved by 7.07%, XGBoost was optimized by 4.31%, and CatBoost achieved an increase of 22.74%. The BO-optimized models possessed more substantial generalization capabilities. The high-quality photon results and the BO-optimized model provided a pathway for accurate supraglacial lake depth retrieval.

C. Performance of the Retrieval Models Before BO

The photon depths from the proposed method train SVR, RF, LightGBM, XGBoost, and CatBoost before BO, and the corresponding results are further analyzed. The MAE, RMSE, and $R^2$of machine learning models are shown in Fig. 14.

Learning machines get the largest MAE and RMSE in Lake 1, and the largest R^2 in Lake 4. The maximum RMSE and MAE are obtained from the results of SVR, where the RMSE is 0.80 m and the MAE is 0.56 m. When processing Lake 4, R^2 obtained by CatBoost is 0.99, which is the largest. The error of retrieval models is small when dealing with Lake 4. Among them, the minimum RMSE obtained by CatBoost in processing Lake 4 is 0.07 m, the minimum MAE obtained by RF in processing Lake 4 is 0.08 m, and the minimum R^2 obtained by LightGBM in processing Lake 1 is 0.73.

It can be found that there is a more significant error during the retrieval of deeper lakes, such as Lake 1. Despite the pronounced undulations of the lakebed in Lake 4, its shallower water depth still ensures the highest accuracy.

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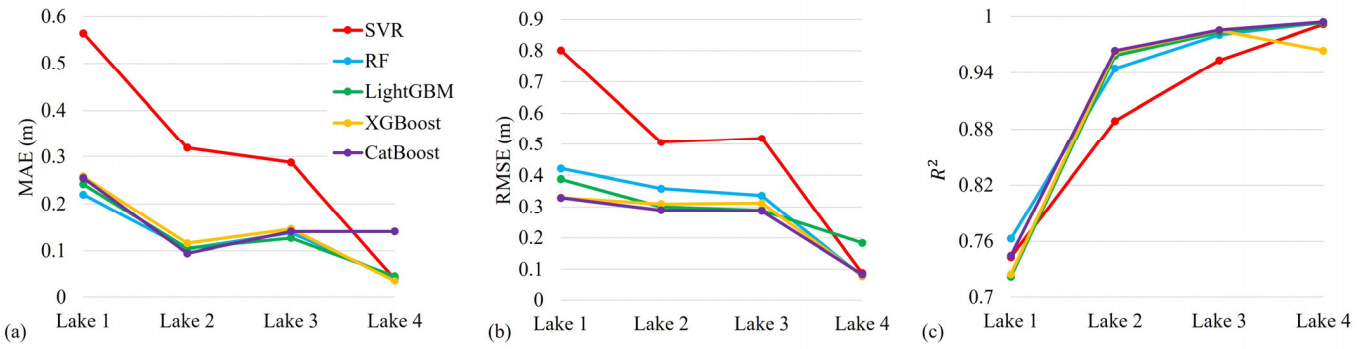


Fig. 14. Accuracy verification results of different machine learning methods before parameter optimization. (a) RMSE, (b) MAEs, (c) R^2s, and (d) MREs for Lakes 1–4. Red line is for SVR, blue line is for RF, green line is for LightGBM, yellow line is for XGBoost, and purple line is for CatBoost.

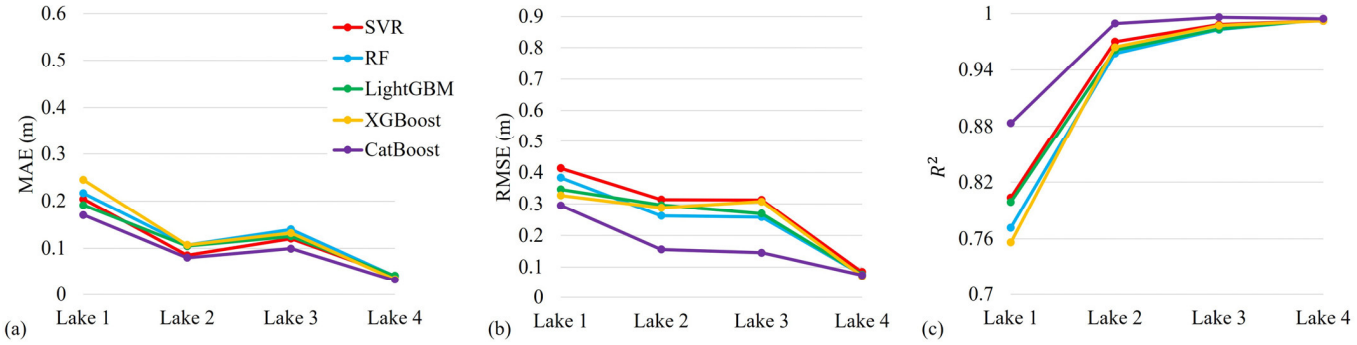


Fig. 15. Accuracy verification results of different machine learning methods after parameter optimization. (a) RMSE, (b) MAEs, (c) R^2s, and (d) MREs for Lakes 1–4. Red line is for SVR, blue line is for RF, green line is for LightGBM, yellow line is for XGBoost, and purple line is for CatBoost.

D. Performance of the Retrieval Models After BO

Previous studies have shown that BO can enhance retrieval performance, but the improvements of the optimized models vary. Fig. 15 illustrates the accuracy of the models in different lakes. Overall, the accuracy for all lakes has improved. BO-XGBoost achieved the highest MAE of 0.24 m for Lake 1. BO-SVR achieved the highest RMSE of 0.41 m for Lake 1. BO-XGBoost also had the worst correlation for Lake 1, with an R^2 of 0.75. In contrast, BO-CatBoost outperformed the other BO-optimized models in MAE, RMSE, and R^2. This conclusion is consistent with the analysis presented earlier.

Although BO has enhanced the model performance, the error patterns of the lakes are still similar to those before parameter optimization. The accuracy for Lake 1 is also the poorest, while that for Lake 4 is also the best. Table III further illustrates the BO improvement in different lakes. For Lake 4, BO-LightGBM achieved the most significant improvement compared with the initial LightGBM, reaching 58.70%. For Lake 1, BO-XGBoost achieved the most minor improvement, at only 0.01%. BO-SVR had the most significant overall improvement, while BO-XGBoost had the smallest. This conclusion is consistent with the analysis presented earlier.

As shown in Figs. 12 and 13, the parameter optimization has demonstrated its advantages in mapping supraglacial lakes. Models optimized by BO can better characterize the nonlinear relationship between photon depth and pixel values, achieving higher accuracy. This makes parameter optimization necessary for accurate depth retrieval of supraglacial lakes.

TABLE III

RELATIVE RMSE IMPROVEMENT IN DIFFERENT LAKES AFTER PARAMETER OPTIMIZATION USING THE LEAVE-ONE-LAKE-OUT PRINCIPLE BY IMPLEMENTING CROSS VALIDATION

Lake	Retrieval model	Relative RMSE improvement (%)
Lake 1	BO-SVR	48.39
	BO-RF	9.02
	BO-LightGBM	10.80
	BO-XGBoost	0.01
	BO-CatBoost	10.18
Lake 2	BO-SVR	38.02
	BO-RF	26.70
	BO-LightGBM	1.27
	BO-XGBoost	7.04
	BO-CatBoost	46.22
Lake 3	BO-SVR	39.77
	BO-RF	23.01
	BO-LightGBM	6.84
	BO-XGBoost	1.54
	BO-CatBoost	49.51
Lake 4	BO-SVR	3.99
	BO-RF	2.82
	BO-LightGBM	58.70
	BO-XGBoost	11.07
	BO-CatBoost	12.60

E. Innovations, Applications, and Limitations

Supraglacial lakes are precursors and concentrators of melting. However, they often appear in areas far from human activities and change rapidly, which restricts researchers' understanding of the location and depth of large-scale supraglacial lakes. After the launch of ICESat-2, researchers

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noticed its performance in depth measurement and proposed a method to measure the lake depth [23], [64]. The lake depth measured by ICESat-2 is concentrated in the satellite trajectories, and it needs to be combined with joint depth retrieval to extend the depth to the whole lake. This article proposes a joint supraglacial lake detection and mapping method using a machine learning method. This method can obtain the lake's position and depth with high accuracy and consistency. In addition, all the processing steps in the proposed method are nonparametric or insensitive to parameters, which can automatically process the data from ICESat-2 and Sentinel-2, reserve the method for large-scale supraglacial lake observation in the whole melting season, and promote supraglacial lake remote sensing to enter the satellite era.

ICESat-2 has been in orbit for nearly five years, providing much polar data. This research undoubtedly aims to perfect and enrich the scientific objectives of ICESat-2. Compared with field measurement, the combined processing of ICESat-2 data and multispectral satellite images can obtain the melting information of supraglacial lakes, reveal the multitime-scale changes and driving factors of glacier melting, and finally, form a comprehensive understanding of the material balance mechanism in Antarctica and Greenland.

However, the satellite remote sensing of supraglacial lakes still has its limitations. Blocking ice on the lake's surface makes it impossible for researchers to measure the supraglacial lake. As shown in Fig. 11(a) and (b), the ice layer is everywhere in lakes, and neither ICESat-2 nor multispectral images can obtain the information below it. When the surface of the supraglacial lake is covered with ice, the proposed joint processing method can only measure the uncovered part of the ice lake not the whole lake. This limitation will cause the lake bed to be described as a frustum and roughly estimated when calculating the volume of supraglacial lakes [99].

The rapid change of the supraglacial lake state requires the data from ICESat-2 and multispectral images to be in the same time phase. The requirement of the multisource data phase limits the development of satellite remote sensing of supraglacial lakes, and finding the same phase data is challenging. In this study, the results of four lakes on AIS were obtained due to the limitation of data. Fortunately, the successful application of few-shot networks to point clouds and images [100], [101] makes the experience gained by learning machines in joint retrieval portable. It can be applied to the remote sensing of the supraglacial lake on images. Compared with the amount of ICESat-2 data, the open-source multispectral satellite images also provide the database for this knowledge transfer.

VII. Conclusion

The lack of continuous survey data of supraglacial lakes in time and space limits researchers' understanding of polar glacier material balance. Based on the existing research, a method to automatically detect and map supraglacial lakes by machine learning is proposed. By this method, the lake depths in Antarctica and Greenland can be obtained only from ICESat-2 and multispectral images. First, a joint lake detection method is proposed, which uses the DI value of the ICESat-2 signal photon to identify the lake photons along the trajectory and extracts the lake pixels from the image according to their $NDWI_{ice}$. Second, a joint lake surveying method is proposed, and the lake photon classification, scattering signal elimination, and refraction correction are carried out on the ICESat-2 data to obtain the lake photon depth. Taking the photon depth and image reflectivity of ICESat-2 as prior information, the learning machines are trained, and finally, the supraglacial lake depth is retrieved. The results on four supraglacial lakes in AIS, Antarctica, show that the accuracy of lake photon filtering reaches 99.21%, and the accuracy of lake region extraction reaches 95.16% with the assistance of photons. The lake depths RMSE and R^2 calculated from ICESat-2 data are 0.26 m and 0.91, respectively. Many retrieval models have good accuracy and consistency in lake depth retrieval, while BO further improves the results. Through self-verification, the RMSE of BO-CatBoost is 0.17 m, and the performance is the best. The proposed method only uses satellite-based remote sensing data, thus removing the dependence on field measurement. In the future, using a large number of data collected by ICESat-2 in Antarctica and Greenland and open-source multispectral satellite images, we can implement the continuous observation of supraglacial lakes in the melting season, especially in remote areas lacking prior observation to understand the glacier mass balance mechanism fully.

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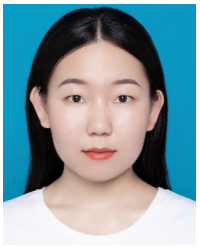
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